University of Waterloo Cryptography Comprehensive Exam Problems and Solutions

September 21, 2026

About this document

This document contains problems and solutions from the University of Waterloo's Cryptography Comprehensive Exam. The problems are organized by year, with each year containing a set of problems and their corresponding solutions. The problems are taken from the official exam papers, and the solutions are written by graduate students at the University of Waterloo.

The Comprehensive Exam Syllabus for Cryptography outlines the topics covered in the exam. Original sources for problems are available in these exam files: 2025, 2020, 2019, 2018, 2017, 2016, 2013, 2011, 2007, 2004, 2003, 2002.

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How to Contribute

Each problem is stored in a single unified file `problems/YYYY/X.tex` (where `YYYY` is the exam year and `X` is the problem number).

To contribute or update a solution:

1. Open the relevant problem file `problems/YYYY/X.tex`.
2. Add or edit the `\begin{solution} …\end{solution}` block after the statement.
3. Update the `% Status:` comment to one of:
   - `complete` — Complete and verified solution.
   - `draft` — Partial or draft solution in progress.
   - `none` — Unsolved / no solution yet.
4. If the problem is identical or almost identical to a problem from another year, add `% Same as: YYYY/X` to group them together.
5. (Optional) Add your name to `% Author:` and to the Contributors list below.

Contributors

The following list contains the names of those who have contributed to this document. Contributors are encouraged to add their names here as co-authors of this work. The list might not be accurate.

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Solution Status Overview

This table shows the current status of solutions for each problem across all exam years. Click on any problem to navigate to its statement in Part I (Problems grouped by topic).

	1	2	3	4	5	6	7	8
2025	HF	SKC	ENT	DS	DL	EC	DL	—
2020	PS	SKC	ENT	RSA	DL	HF	DS	—
2019	ENT	HF	SKC	RSA	EC	PS	—	—
2018	SKC	HF	ENT	ENT	RSA	EC	DS	—
2017	SKC	HF	ENT	RSA	DS	DL	EC	—
2016	HF	ENT	ENT	RSA	DL	SKC	DS	PS
2013	HF	SKC	ENT	PS	RSA	EC	—	—
2011	HF	SKC	ENT	PS	SKC	EC	—	—
2007	HF	ENT	DL	DL	EC	EC	—	—
2004	HF	ENT	DL	DS	PS	EC	—	—
2003	SKC	ENT	RSA	DL	DL	EC	—	—
2002	SKC	HF	ENT	RSA	DL	DS	DS	—

Table of Contents

Part I: Problems by topic

Elementary Number Theory ..... p. 6
Discrete Logarithm ..... p. 10
Elliptic Curves ..... p. 14
Provable Security ..... p. 16
Digital Signatures ..... p. 19
Symmetric-Key Cryptography .... p. 24
Hash Functions ..... p. 29
RSA Cryptosystem ..... p. 34

Part II: Solutions by topic

Elementary Number Theory ..... p. 38

Part III: Problems by year

2025 Exam ..... p. 50
2020 Exam ..... p. 53
2019 Exam ..... p. 56
2018 Exam ..... p. 59
2017 Exam ..... p. 62
2016 Exam ..... p. 65
2013 Exam ..... p. 68
2011 Exam ..... p. 71
2007 Exam ..... p. 74
2004 Exam ..... p. 77
2003 Exam ..... p. 80
2002 Exam ..... p. 83

Part I.

Problems grouped by topic

Elementary Number Theory

Problem 3/2025, 3/2020: Primality testing (solution).

Recall that if n is a composite integer, and $a \in [1, n-1]$ satisfies $a^{n-1} \equiv 1 \pmod{n}$, then a is called a Fermat liar for n.

Now, let $n = pq$, where p and q are distinct odd primes, let $a \in \mathbb{Z}_n^\times$, and let $d = \gcd(p-1, q-1)$.

(a) Prove that a is a Fermat liar for n if and only if $a^d \equiv 1 \pmod{n}$.

(Hint: Note that $n - 1 = pq - 1 = (p-1)q + (q-1)$.)

(b) Suppose now that $q = 2p + 1$. How many $a \in \mathbb{Z}_n^*$ are Fermat liars for n? List them all (in terms of p).

Problem 1/2019: Elementary number theory (solution).

Let $n = pq$ be an RSA modulus. Prove the following:

$$\prod_{x \in \mathbb{Z}_n^*} x \equiv 1 \pmod{n}.$$

Problem 3/2018: Elementary number theory (solution).

(a) Let $n \geq 3$ be an integer. Suppose that there exists an integer a such that $a^{n-1} \equiv 1 \pmod{n}$ and $a^{(n-1)/q} \not\equiv 1 \pmod{n}$ for all prime divisors q of $n - 1$. Prove that n is prime.

(b) The *Fermat numbers* are $F_k = 2^{2^k} + 1$ for $k \geq 1$. Prove that for $k \geq 2$, F_k is prime if and only if $5^{(F_k-1)/2} \equiv -1 \pmod{F_k}$. (It may help to remember Euler's Theorem: If p is an odd prime, then $\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}$.)

Problem 4/2018: Number-theoretic algorithms.

(a) An instance of CHREM (Chinese Remainder Problem) is a pair of distinct primes p and q, and two integers $a \in [0, p-1]$ and $b \in [0, q-1]$. The problem is to determine the unique integer $x \in [0, pq-1]$ such that $x \equiv a \pmod{p}$ and $x \equiv b \pmod{q}$. Design (and analyze) a polynomial-time algorithm for CHREM.

(b) Let $n = pq$, where p and q are distinct primes satisfying $p \equiv q \equiv 3 \pmod{4}$. Let FACTOR be the problem of factoring n. Let SQUARE-ROOT be the problem of finding one square root of a $\in \mathbb{Z}_n$ that is a quadratic residue modulo n. Prove that SQUARE-ROOT $\leq_P$ FACTOR.

Problem 3/2017: Elementary number theory (incomplete solution).

Note: Parts (a) and (b) are unrelated.

(a) Let p be a prime, $n \in \mathbb{N}$, and $q = p^n$. Prove that the finite field $\mathbb{F}_q$ has $q-2$ generators if and only if $q-1$ is a Mersenne prime.

(b) Let $m \geq 3$ be an integer. Prove that if a is a quadratic residue modulo m, and $ab \equiv 1 \pmod{m}$, then b is also a quadratic residue. Now let p be a prime of the form $p = 4k+3$. Prove that the product of all the quadratic residues modulo p is congruent to 1.

Problem 2/2016, 2/2003: Fermat primes and generators (solution).

Let $p = 2^m + 1$ be a prime, with $m \geq 1$.

(a) Prove that m is a power of 2.

(b) Prove that any quadratic nonresidue modulo p is a generator of $\mathbb{F}_p^\times$.

(c) Prove that 5 is a generator of $\mathbb{F}_p^\times$, except in the case $p = 5$.

(d) Prove that 7 is a generator of $\mathbb{F}_p^\times$, except in the case $p = 3$.

Problem 3/2016: Elementary number theory (solution).

Let n be an RSA modulus. Does n always, sometimes, or never have a primitive root? (Recall that a primitive root modulo n is an element of order $\phi(n)$ in the multiplicative group of units $\mathbb{Z}_n^*$.)

Problem 3/2013: Elementary number theory (incomplete solution).

Let p denote an odd prime.

(a) Let q be a generator (a.k.a. a primitive root) modulo p. Show that q is not a quadratic residue modulo p.

(b) Suppose $y \equiv g^x \pmod{p}$ for some integer $0 < x < p-1$. Show how one can efficiently find the least significant bit of the binary expansion of x.

Problem 3/2011: Elementary number theory.

Throughout this question, n is an odd integer, and a is an integer, $1 < a < n$.

(a) Assume n is prime, and write $n - 1 = 2^k m$ with m odd. Assume $a^{2^i m} \not\equiv -1 \pmod{n}$ for all $0 \leq i < k$. Prove that $a^{2^i m} \equiv 1 \pmod{n}$ for all $0 \leq i \leq k$.

(b) Let n be prime. Prove that $\left(\frac{a}{n}\right) \equiv a^{(n-1)/2} \pmod{n}$.

(c) Now let n be composite, again write $n - 1 = 2^k m$ with m odd. Assume $\gcd(a, n) = 1$. Recall that n is called a strong pseudoprime to the base a if

$$a^n \equiv 1 \pmod{n} \quad \text{or} \quad \text{there exists } 0 \leq i < k \text{ such that } a^{2^i m} \equiv -1 \pmod{n}.$$

Further recall that n is called an Euler pseudoprime to the base a if

$$a^{(n-1)/2} \equiv \left(\frac{a}{n}\right) \pmod{n}.$$

Prove that for $n \equiv 3 \pmod{4}$, n is a strong pseudoprime to the base a if and only if n is an Euler pseudoprime to the base a.

Problem 2/2007: Elementary number theory (solution).

(a) Let p be an odd prime. Prove that a quadratic residue modulo p can never be a generator of $\mathbb{F}_p^*$.

(b) Let p be an odd prime. Prove that -3 is a quadratic residue modulo p if and only if $p \equiv 1 \pmod{3}$.

(c) Let $p > 3$ be a Mersenne prime. Prove that 3 is a quadratic nonresidue modulo p.

Problem 2/2004: Elementary number theory.

Let p be an odd prime and let $n > 1$ be a positive integer. Recall that the multiplicative group $\mathbb{Z}_{p^n}^*$ is cyclic. Suppose that the integer g is a generator of $\mathbb{Z}_p^*$, and let $h = (p+1)g$. Prove that at least one of g or h is a generator of $\mathbb{Z}_{p^n}^*$. Hint: First prove that at least one of g^{p-1} or h^{p-1} can be expressed in the form $1 + kp$ where k is an integer that is not divisible by p.

Problem 3/2002: Carmichael numbers (solution).

Assume throughout this question that n is square-free (i.e., n is not divisible by the square of a prime). Then $n = p_1 \cdots p_\ell$, where $p_1, \ldots, p_\ell$ are distinct primes. Such an integer n is a Carmichael number if $a^{n-1} \equiv 1 \pmod{n}$ for all integers a that are relatively prime to n. You may use the following facts in your solutions: (i) $a^{n-1} \equiv 1 \pmod{p_i}$ for all i implies $a^{n-1} \equiv 1 \pmod{n}$; (ii) for every prime p, there exists a primitive element mod p.

(a) Suppose that n is a Carmichael number. Prove that $p_i - 1$ divides $n - 1$ for all $i, 1 \leq i \leq \ell$.

(b) Suppose that $p_i - 1$ divides $n - 1$ for all $i, 1 \leq i \leq \ell$. Then prove that n is a Carmichael number.

(c) Prove that 561 is a Carmichael number.

Discrete Logarithm

Problem 5/2025: Discrete logarithms.

Let p be a prime, let q be a prime divisor of $p-1$, and let g be an element of order q in $\mathbb{Z}_p^*$. The discrete logarithm problem is the following: given p, q, g and $h \in_R \langle g \rangle$, determine the integer $x \in [0, n-1]$ such that $h = g^x \pmod{p}$. Here, $\langle g \rangle$ denotes the subgroup of $\mathbb{Z}_p^*$ generated by g.

Describe and analyze an algorithm for solving the discrete logarithm problem that has running time $O(\sqrt{q})$ $\mathbb{Z}_p$ operations. There is no limit on how much space your algorithm can use.

Problem 7/2025: Elliptic curve Diffie-Hellman key agreement.

Let E be an elliptic curve defined over $\mathbb{Z}_p$, where p is a prime. Let $n = \#E(\mathbb{Z}_p)$, and suppose that n is prime. Let $P \in E(\mathbb{Z}_p)$ with $P \neq \infty$.

(a) Recall that in the unauthenticated elliptic curve Diffie-Hellman key agreement protocol (ECDH), Alice selects $x \in_R [1, n-1]$ and sends $X = xP$ to Bob. Similarly, Bob selects $y \in_R [1, n-1]$ and sends $Y = yP$ to Alice. Their shared secret key is $k = H(K)$ where $K = xyP$ and H is a hash function. Alice and Bob subsequently use k to encrypt and decrypt messages using AES.

Describe the malicious-intruder-in-the-middle attack (also known as the man-in-the-middle attack) on ECDH.

(b) Describe a variant of authenticated ECDH, and give an informal argument for its security against active attacks.

Problem 5/2020, 6/2017: Discrete logarithm and Diffie-Hellman problems.

Let G be a cyclic group of prime order $n > 2$ generated by α. The notation $A \leq_P B$ means that problem A polynomial-time reduces to problem B.

(a) Recall that the discrete logarithm problem in G with respect to α (DLP_α) is the following: given $\gamma \in G$, find the integer $\ell \in [0, n-1]$ that satisfies $\gamma = \alpha^\ell$. Now, let β be another generator of G. Prove that $\text{DLP}_\alpha \leq_P \text{DLP}_\beta$. (This proves that the hardness of the DLP does not depend on the choice of generator.)

(b) Recall that the Diffie-Hellman Problem (DHP) is the following: given $\alpha^x, \alpha^y \in G$, compute α^{xy}. The problem INV is the following: given $\alpha^x \in G$, compute $\alpha^{x^{-1}}$ (where x^{-1} is the inverse of x modulo n). Prove that $\text{INV} \leq_P \text{DHP}$.

(c) The problem SQUARE is the following: given $\alpha^x \in G$, compute α^{x^2}. Prove that $\text{DHP} \leq_P \text{SQUARE}$.

(d) The problem CUBE is the following: given $\alpha^x \in G$, compute α^{x^3}. Prove that $\text{CUBE} \leq_P \text{SQUARE}$.

(e) Prove that $\text{DHP} \leq_P \text{CUBE}$. (Hint: Recall that $(x+1)^3 = x^3 + 3x^2 + 3x + 1$.)

Problem 5/2016, 3/2007: Discrete logarithm problem.

Let $p = 2^k + 1$ be a prime number. Describe and analyze a polynomial-time algorithm for solving the discrete logarithm problem in $\mathbb{Z}_p^*$. (Recall that the DLP in $\mathbb{Z}_p^*$ is the following: given p, a generator g of $\mathbb{Z}_p^*$, and $h \in \mathbb{Z}_p^*$, find the integer $\ell \in [0, p-2]$ such that $h = g^\ell \bmod p$.)

Problem 4/2007, 5/2003: Security of the basic ElGamal public-key encryption scheme.

Let G be a (cyclic) group of prime order $n > 2$, and let α be a generator of G. We assume that the group operation in G can be computed in polynomial time. Recall that the Diffie-Hellman problem for G with respect to α (DHP_α) is the following: given α^a and α^b, compute α^{ab}. The Decisional Diffie-Hellman problem for G with respect to α (DDHP_α) is the following: given α^a, α^b, and α^c, decide whether $c \equiv ab \pmod{n}$.

In the basic ElGamal public-key encryption scheme, Alice's private key is an integer $a \in [1, n-1]$, and her public key is $\beta = \alpha^a$. To encrypt a plaintext message $m \in G$ for Alice, Bob selects $k \in_R [1, n-1]$, and sends the ciphertext $C = (\alpha^k, m \cdot \beta^k)$ to Alice.

In the following, we consider ciphertext-only attacks on the basic ElGamal public-key encryption scheme. The attacker has knowledge of the group parameters, Alice's public key β, and one or more ciphertexts.

(a) The ElGamal-decrypt problem is the following: Given a public key β and a ciphertext C, compute the corresponding plaintext. Prove that the ElGamal-decrypt problem is polynomial-time equivalent to DHP_α.

(b) Prove that the semantic security of the basic ElGamal public-key encryption scheme (under ciphertext-only attack) is polynomial-time equivalent to DDHP_α.

(c) Is the basic ElGamal public-key encryption scheme semantically secure against chosen-plaintext attacks? (Justify your answer.)

Problem 3/2004: Bit security of the discrete logarithm problem.

Let p be an odd prime, and let g be a generator of $\mathbb{Z}_p^*$. Consider the following three problems: DLP: Given p, g, and $x \in \mathbb{Z}_p^*$, determine the integer $a \in [0, p-2]$ such that $x \equiv g^a \pmod{p}$. (We write $a = \log_g x$.) DLP-LSB: Given p, g, and $x \in \mathbb{Z}_p^*$, determine $A(x)$ where

$$A(x) = \begin{cases} 1, & \text{if } \log_g x \text{ is even,} \\ 0, & \text{if } \log_g x \text{ is odd.} \end{cases}$$

DLP-MSB: Given p, g, and $x \in \mathbb{Z}_p^*$, determine $B(x)$ where

$$B(x) = \begin{cases} 1, & 0 \le \log_g x < \frac{p-1}{2}, \\ 0, & \frac{p-1}{2} \le \log_g x \le p-2. \end{cases}$$

(a) Prove that DLP $\le_p$ DLP. (Recall that A $\le_p$ B means that problem A polynomial-time reduces to problem B.) (b) Prove that DLP $\le_p$ DLP-MSB. (c) Does DLP $\le_p$ DLP-LSB? (Justify your answer.)

Problem 4/2003: Diffie-Hellman problem.

This exercise shows that the hardness of the Diffie-Hellman problem does not depend on the choice of generator.

Let G be a (cyclic) group of prime order $n > 2$, and let α be a generator of G. We assume that the group operation in G can be computed in polynomial time. Recall that the Diffie-Hellman problem for G with respect to α (DHP_α) is the following: given α^a and α^b, compute α^{ab}. In this question, you are given a polynomial-time algorithm A which solves DHP_α.

(a) Devise a polynomial-time algorithm which on input α^a and a positive integer k, outputs α^{a^k}.

(b) Devise a polynomial-time algorithm which on input α^a (with $a \not\equiv 1 \pmod{n}$), outputs $\alpha^{a^{-1}}$.

(c) Let β be a generator of G. Devise a polynomial-time algorithm for solving DHP_β (i.e., given β^a and β^b, compute β^{ab}).

Problem 5/2002: Bit security of the discrete logarithm problem.

Let p be a prime with $p \equiv 3 \pmod{4}$. Let $\alpha \in \mathbb{Z}_p^*$ be a generator of $\mathbb{Z}_p^*$. The discrete logarithm problem in $\mathbb{Z}_p^*$ is the following: given α and $\beta \in \mathbb{Z}_p^*$, find the integer a, $0 \leq a \leq p-2$, such that $\beta = \alpha^a \pmod{p}$. Let $L_1(\beta)$ denote the least significant bit of a. That is, $L_1(\beta) = 0$ if a is even, and $L_1(\beta) = 1$ if a is odd. Let $L_2(\beta)$ denote the second least significant bit of a. That is, $L_2(\beta) = 0$ if $a \equiv 0$ or $1 \pmod{4}$, and $L_2(\beta) = 1$ if $a \equiv 2$ or $3 \pmod{4}$.

(a) Let $\gamma \in \mathbb{Z}_p^*$ be a quadratic residue modulo p. Show that the two square roots of γ modulo p are $\pm\gamma^{(p+1)/4}$.

(b) Show that $L_1(\beta)$ can be efficiently computed given p, α, β.

(c) Prove that $L_1(\beta) \neq L_1(\beta^{-1})$.

(d) Suppose that you have an efficient algorithm A (an oracle) for computing $L_2(\beta)$ given p, α, β. Use A to devise an efficient algorithm for computing $L_1(\beta)$ and hence break the discrete logarithm problem. Briefly justify that your algorithm is correct and efficient.

Elliptic Curves

Problem 6/2025, 6/2018, 6/2013, 6/2003: Elliptic curves.

Let q be a power of an odd prime satisfying $q \equiv 2 \pmod 3$. Consider the elliptic curve $E\colon y^2 = x^3 + b$ defined over $\mathbb{F}_q$ ($b \neq 0$).

(a) Prove that the mapping $x \mapsto x^3$ is a bijection on $\mathbb{F}_q$.

(b) Prove that the number of points in $E(\mathbb{F}_q)$ is $q + 1$.

(c) Prove that $E(\mathbb{F}_q)$ is cyclic.

(d) Let $R = (x_R, y_R)$ be a point in $E(\mathbb{F}_q)$. Given y_R, explain how to compute x_R efficiently.

Problem 5/2019, 7/2017: Elliptic curves.

Let $E\colon y^2 = x^3 + ax + b$ be an elliptic curve defined over $\mathbb{Z}_p$, where $p > 3$ is prime.

(a) Prove the formula

$$\#E(\mathbb{Z}_p) = p + 1 + \sum_{x=0}^{p-1} \left(\frac{x^3 + ax + b}{p} \right),$$

where the expression inside the summation is the Legendre symbol.

(b) Suppose that $x^3 + ax + b$ splits into three distinct linear factors modulo p. Show that $E(\mathbb{Z}_p)$ is not cyclic.

(c) If furthermore $b = 0$ and $p \equiv 3 \pmod 4$, show that $\#E(\mathbb{Z}_p) = p + 1$.

Problem 6/2011: Elliptic curves.

Consider the Kobitz (Koblitz) curve defined over $\mathbb{F}_2$:

$$E:\ y^2 + xy = x^3 + x^2 + 1.$$

(a) Show that $\#E(\mathbb{F}_2^r)$ is even for all $r \geq 1$.

(b) Show that if $r > 4$ and $\#E(\mathbb{F}_{2^r}) = 2p$ where p is a prime number, then r must be prime.

Problem 5/2007: Weil pairing.

Let E be an elliptic curve over a field, and suppose $P, Q \in E[n]$ for some $n > 0$.

(a) Give the formula for the Weil pairing $e(P, Q)$ of P and Q.

(b) Prove that the choice of divisors in the formula does not affect the value of $e(P, Q)$. You may assume Weil reciprocity.

(c) Prove that the Weil pairing is bilinear.

Problem 6/2007: Embedding degree.

Let E be an elliptic curve defined over $\mathbb{F}_q$, and denote by n the largest prime factor of $\#E(\mathbb{F}_q)$. Recall that the embedding degree of E is by definition equal to the multiplicative order of q in $(\mathbb{Z}/n\mathbb{Z})^*$.

(a) Let k be an integer. Prove that the embedding degree of E divides k if and only if $n \mid (q^k - 1)$, where t is the trace of E.

(b) Suppose $q = p^2$ and $t = p$ where p is prime. Determine the embedding degree of E.

Problem 6/2004: Elliptic curves and finite fields.

(a) Recall that the Trace function $\mathrm{Tr} : \mathbb{F}_{2^m} \to \mathbb{F}_2$ is defined by $\mathrm{Tr}(\alpha) = \sum_{i=0}^{m-1} \alpha^{2^i}$. Prove that exactly half of the elements in $\mathbb{F}_{2^m}$ have trace 0 (and the other half have trace 1). (b) Let $\alpha \in \mathbb{F}_{2^m}$. Prove that the equation $x^2 + x = \alpha$ has a solution $x \in \mathbb{F}_{2^m}$ if and only if $\mathrm{Tr}(\alpha) = 0$. (c) Let $E : y^2 + y = x^3$ be an elliptic curve over $\mathbb{F}_{2^m}$ where m is an odd positive integer. Prove that $|E(\mathbb{F}_{2^m})| = 2^m + 1$.

Provable Security

Problem 1/2020: Shannon's theory.

(a) Define what it means for a symmetric-key encryption scheme to have perfect secrecy.

(b) Suppose that a symmetric-key encryption scheme with encryption function $E\colon \mathcal{K} \times \mathcal{M} \to \mathcal{C}$ satisfies $|\mathcal{K}| = |\mathcal{C}| = |\mathcal{M}|$. Prove that, if the encryption scheme has perfect secrecy, then every key is used with equal probability $1/|\mathcal{K}|$, and for every message $x \in \mathcal{M}$ and ciphertext $y \in \mathcal{C}$, there is a unique key k such that $E(k, x) = y$.

Problem 6/2019: Provable security.

The Boneh–Franklin cryptosystem is defined as follows.

Public parameters: A cryptographic pairing $e : G \times G \to GT$, and two elements $g, h \in G$, each of which generates the cyclic group G.

Key generation: Choose $\alpha \in \mathbb{Z}$. The public key is g^α. The private key is h^α.

Encryption: Given $m \in \{0,1\}^k$, the encryption of m is $(g^r, e(g^\alpha, h^r) \oplus m)$ for random $r \in \mathbb{Z}$. Here k is the number of bits used to represent an element of GT.

Decryption: Given a ciphertext (c_1, c_2), the plaintext is $c_2 \oplus e(c_1, h^\alpha)$.

(a) Show that the Boneh–Franklin scheme is **OW-CPA** under the Bilinear Diffie–Hellman (BDH) assumption: Given g, g^α, g^β, and h, it is infeasible to compute $e(g^\alpha, h^\beta)$ from the ciphertext.

(b) Show that the Boneh–Franklin scheme is **IND-CPA** under the Decisional Bilinear Diffie–Hellman (DBDH) assumption.

$\text{OW-CPA}^{\mathcal{A}}_{\text{Enc}}$	
1:	$(\mathsf{pk}, \mathsf{sk}) \leftarrow_{\$} \mathsf{KGen}(1^\lambda)$
2:	$m \leftarrow_{\$} M$
3:	$c \leftarrow_{\$} \mathsf{Enc}(\mathsf{pk}, m)$
4:	$m' \leftarrow_{\$} \mathcal{A}(1^\lambda, \mathsf{pk}, c)$
5:	**return** $m \stackrel{?}{=} m'$

$\text{IND-CPA}^{\mathcal{A}}_{\text{Enc}}$	
1:	$(\mathsf{pk}, \mathsf{sk}) \leftarrow_{\$} \mathsf{KGen}(1^\lambda)$
2:	$b \leftarrow_{\$} \{0,1\}$
3:	$(m_0, m_1) \leftarrow_{\$} \mathcal{A}(1^\lambda, \mathsf{pk})$
4:	$c \leftarrow_{\$} \mathsf{Enc}(\mathsf{pk}, m_b)$
5:	$b' \leftarrow_{\$} \mathcal{A}(1^\lambda, \mathsf{pk}, c)$
6:	**return** $b \stackrel{?}{=} b'$

Figure: OW-CPA and IND-CPA definitions.

Problem 8/2016: Provable security.

Consider the Zheng-Sebberry public-key encryption scheme (1993):

Public parameters: A cyclic group G, a generator g of G, and two random oracles

$$H_1 : \{0,1\}^t \to \{0,1\}^n, \qquad H_2 : G \to \{0,1\}^{t+n}.$$

Key generation: Choose a private key $x \in \mathbb{Z}$. The corresponding public key is $h = g^x$.

Encryption: To encrypt $m \in \{0,1\}^t$, choose $y \in \mathbb{Z}$ and compute

$$Y = g^y, \qquad c = H_2(h^y) \oplus (m \,\|\, H_1(m)).$$

The ciphertext is (Y, c).

Decryption: Compute $c \oplus H_2(Y^x)$. If the leftmost t bits of the result map to the rightmost n bits under H_1, then output the leftmost t bits; otherwise output NULL.

Show that the Zheng-Sebberry scheme is not IND-CCA2. (IND-CCA2 means "indistinguishable against adaptive chosen-ciphertext attack". In this attack, the adversary selects two plaintexts m_0, m_1, is then given the encryption c of m_b (where $b \in \{0,1\}$), and has to determine b with probability significantly greater than $1/2$. The adversary is also given access to a decryption oracle to which it can present any ciphertext for decryption except for the challenge ciphertext c itself.)

Problem 4/2013: Provable security.

Let p be a large prime, and let q be a large prime divisor of $p-1$. Let q be an element of order q in $\mathbb{Z}_p^*$, and let G denote the subgroup of $\mathbb{Z}_p^*$ generated by g. We consider the ElGamal encryption scheme in the group G, which is defined as follows:

Setup: Public parameters p, q, g are chosen.

Key generation: Choose $x \in \mathbb{Z}_q^*$ at random. The public key is g^x and the private key is x.

Encryption: The message $m \in G$ and a public key g^x, choose a random integer r and output the ciphertext $c = (g^x, mg^{xr})$.

Decryption: Given a ciphertext $c = (\rho, \sigma)$, output $m = \sigma/\rho^x$.

(a) It is conjectured that the Decision Diffie-Hellman (DDH) assumption holds in the group G. Assuming this conjecture, prove that the ElGamal encryption scheme is IND-CPA secure. (Recall that IND-CPA means "indistinguishable against chosen-plaintext attacks".)

(b) Suppose that, in the encryption function defined above, the message space is taken to be $\mathbb{Z}_p^*$ instead of G. Show that the resulting encryption scheme is not IND-CPA secure.

Problem 4/2011: Provable security.

The Blum-Goldwasser public key cryptosystem is given as follows.

- Key generation: Choose distinct primes p and q congruent to 3 (mod 4). The public key is $n = pq$ and the private key is (p, q). - Encryption: A message m consists of a single bit. To encrypt m, choose a random $x \in (\mathbb{Z}/n\mathbb{Z})^{\times 2}$ (the quadratic residues modulo n) and compute

$$i)\ b = \mathrm{LSB}(x), \quad ii)\ c = b \oplus m, \quad iii)\ y = x^2 \pmod{n}.$$

The ciphertext is (c, y).

Prove that the cryptosystem is IND-CPA. State any necessary assumptions.

Digital Signatures

Problem 4/2025, 5/2017, 5/2004: RSA signatures.

Recall that in the Full-Domain Hash (FDH) RSA signature scheme, an entity (Alice) with public key (n, e) and private key d generates a signature s on a message m by computing

$$s \equiv H(m)^d \pmod{n}.$$

Here $H\colon \{0,1\}^* \to \{0, 1, \ldots, n-1\}$ is a hash function.

(a) Describe how someone can verify Alice's signature s on a message m.

(b) Show that FDH RSA is insecure against passive adversaries if H is not preimage resistant.

(c) Prove that if finding e-th roots modulo n is intractable, and if H is a random function, then FDH RSA is secure (i.e., existentially unforgeable by an adversary who can mount an adaptive chosen-message attack).

Problem 7/2020: Signature schemes.

(a) Define what it means for a signature scheme to be secure.

In the basic (textbook) RSA signature scheme, the public key is a pair (n, e) and the secret key is a pair (n, d). The signature on a message $m \in \mathbb{Z}_n$ is $s \equiv m^d \pmod{n}$, and verification is done by checking if $s^e \equiv m \pmod{n}$.

(b) Show how to come up with some valid message/signature pairs under a key-only attack on the basic RSA signature scheme.

(c) Show how to come with a forgery for any given message m under a chosen-message attack on the basic RSA signature scheme, wherein the adversary is allowed to obtain signatures for at most two messages (neither of which is equal to m).

(d) One way of building a secure RSA-based signature scheme is to sign $H(m)$ using a "full-domain hash function H". The security proof for RSA-FDH is "in the random oracle model". Briefly explain what the random oracle model is.

Problem 7/2018: ECDSA.

Recall the ECDSA signature scheme. The domain parameters consist of a 256-bit prime p, an elliptic curve E defined over $\mathbb{Z}_p$ with prime $n = \#E(\mathbb{Z}_p)$, and a point $P \in E(\mathbb{Z}_p)$ with $P \neq \infty$. Alice's private key is $a \in_R [1, n-1]$ and her public key is $A = aP$. To sign a message $M \in \{0,1\}^*$, Alice does the following:

(i) Select a per-message secret $k \in_R [1, n-1]$.
(ii) Compute $m = \text{SHA256}(M)$.
(iii) Compute $R = kP$. Let $r = x(R) \bmod n$ and check that $r \neq 0$.
(iv) Compute $s = k^{-1}(m + ar) \bmod n$, and check that $s \neq 0$.
(v) Alice's signature on M is (r, s).

To verify Alice's signature (r, s) on M, Bob does the following:

(i) Obtain an authentic copy of Alice's public key A.
(ii) Check that $1 \leq r, s \leq n-1$.
(iii) Compute $m = \text{SHA256}(M)$.
(iv) Compute $u_1 = ms^{-1} \bmod n$ and $u_2 = rs^{-1} \bmod n$.
(v) Compute $V = u_1 P + u_2 A$ and let $v = x(V) \bmod n$.
(vi) Accept if and only if $v = r$.

(a) Define what it means for a signature scheme to be secure.

(b) Suppose now that an adversary knows a message M such that $\text{SHA256}(M) = 0$. Show that the adversary can efficiently compute a valid signature for M. (The adversary knows the domain parameters and Alice's public key A, but does not have access to a signing oracle.)

Problem 7/2016: Identification schemes.

Let G be a cyclic group of prime order p with generator g. Suppose the verifier is given $\beta = g^\alpha$ for some randomly selected $\alpha \in \mathbb{Z}_p$. Consider the zero-knowledge proof of knowledge of α in Figure 1.

Figure 1: Zero-knowledge proof of knowledge of α

(a) Show that the proof is zero-knowledge for an honest verifier. (b) Show that a cheating prover can succeed with probability 1/2. (c) Describe how to modify the protocol so that the prover's cheating probability is reduced to negligible levels.

Problem 4/2004: Fault analysis attack on the RSA signature scheme.

Suppose that a smart card is using the Chinese Remainder Theorem for RSA signature generation. That is, if (n, e) is the RSA public key and d is the corresponding private key, then signing a message m is performed as follows:

i) Compute $M = H(m)$.

ii) Compute $s_p \equiv M^{d_p} \pmod{p}$ and $s_q \equiv M^{d_q} \pmod{q}$, where $d_p = d \bmod (p-1)$ and $d_q = d \bmod (q-1)$.

iii) Find s, $0 \leq s \leq n-1$, such that

$$s \equiv s_p \pmod{p}, \qquad s \equiv s_q \pmod{q}.$$

(a) Prove that s is the correct signature of m (that is, prove that $s \equiv H(m)^d \pmod{n}$). (b) Explain why it might be advantageous to compute s using the procedure described above instead of computing $s = H(m)^d \bmod n$ directly using the repeated square-and-multiply algorithm. (c) Suppose now that an adversary can somehow induce the smart card to compute s_p incorrectly (and s_q correctly) while signing a message. Let s' be a resulting (incorrect) signature on m. Suppose that the adversary has access to the public key (n, e) and also the signed message (m, s'). Show how the adversary can efficiently factor n. (d) Suggest a (realistic and practical) method for preventing this attack.

Problem 6/2002: Hash functions and DSA.

We recall the DSA signature scheme. The system parameters consist of a 1024-bit prime p, a 160-bit prime divisor q of $p-1$, and an element $g \in \mathbb{Z}_p^*$ of order q. SHA-1 is a 160-bit hash function. Alice's private key is $a \in \mathbb{Z}_p^*$, while her public key is $h = g^a \bmod p$. To sign a message $M \in \{0,1\}^*$, Alice does the following:

1) Choose a per-message secret number k with $1 \leq k \leq q-1$ and compute $r \equiv (g^k \bmod p) \bmod q$. If $r = 0$, choose another k.

2) Compute $s \equiv k^{-1}(\mathrm{SHA}-1(M) + ar) \pmod{q}$. If $s = 0$, choose another k.

The signature on M is the pair (r, s).

To verify a signature (r, s) on M, Bob does the following:

(i) Obtain an authentic copy of Alice's public key h.

(ii) Compute $m = \mathrm{SHA}-1(M)$.

(iii) Check that $1 \leq r, s \leq q-1$.

(iv) Compute $u_1 = ms^{-1} \bmod q$ and $u_2 = rs^{-1} \bmod q$.

(v) Accept if $r \equiv (g^{u_1} h^{u_2}) \bmod p \bmod q$. (Otherwise reject.)

Recall that a signature scheme is secure if it is existentially unforgeable by chosen-message attacks. It is insecure if it is not secure.

(a) Define what it means for SHA-1 to be preimage resistant.

(b) Define what it means for SHA-1 to be 2nd preimage resistant.

(c) Define what it means for SHA-1 to be collision resistant.

(d) Prove that DSA is insecure if SHA-1 is not preimage resistant.

(e) Prove that DSA is insecure if SHA-1 is not 2nd preimage resistant.

(f) Prove that DSA is insecure if SHA-1 is not collision resistant.

Problem 7/2002: Duplicate signatures.

The Elliptic Curve Digital Signature Algorithm (ECDSA) is as follows: Let p be a prime, and let E be an elliptic curve defined over $\mathbb{F}_p$. Let A be a point on E having prime order q, such that the Discrete Logarithm problem in $\langle A \rangle$ is infeasible. Let $\mathcal{P} = \{0,1\}^*$ denote the set of messages. Define the public parameters $\mathcal{K} = \{(p, q, E, A, m, B) : B = mA\}$, where $0 \leq m \leq q-1$. The values p, q, E, A and B are the public key, and m is the private key.

The signature for a message $x \in \mathcal{P}$ is computed as follows: For $K = (p, q, E, A, m, B)$, and for a (secret) random number k, $1 \leq k \leq q-1$, define

$$sig_K(x, k) = (r, s),$$

where $kA = (u, v)$ and $r = u \bmod q$, $s = k^{-1}(\mathrm{SHA}-1(x) + mr) \bmod q$.

(If either $r = 0$ or $s = 0$, a new random value of k should be chosen.)

To verify A's signature (r, s) on M, Bob does the following:

(i) Obtain an authentic copy of Alice's public key h.

(ii) Compute $m = \mathrm{SHA}-1(M)$.

(iii) Check that $1 \leq r, s \leq q-1$.

(iv) Compute $u_1 = ms^{-1} \bmod q$ and $u_2 = rs^{-1} \bmod q$.

(v) Accept iff $r = (hu_1 + Bu_2)\restriction_q$ (i.e., the x-coordinate modulo q of the point $u_1 A + u_2 B$).

Recall that a signature scheme is secure if it is existentially forgeable, etc.

(a) Suppose that two messages x and x' are signed with the same k under the same key, producing signatures (r, s) and (r', s'). Show that if SHA-1 is used as the hash, then $x \neq x'$ can still yield the same signature (r, s) with non-negligible probability. Explain how this leads to an attack on the ECDSA signature scheme.

(b) Suppose that two messages x and x' are said to have duplicate ECDSA signatures if SHA-1$(x) \neq$ SHA-1(x') but $\text{sig}_K(x, k) = \text{sig}_K(x', k')$ for some integers k, k' for some key parameters. Suppose the public key parameters p, q, E and A are fixed. Given any two messages x and x' such that SHA-1$(x) \neq$ SHA-1(x'), show that it is possible for Alice to choose a private key m (and hence a corresponding public key B) so that x and x' have duplicate ECDSA signatures under the key p, q, E, A, m and B.

(c) Suppose that Alice, say, signs message x with signature (r, s), and then later claims that she really signed the message x', where (r, s) is also a signature on x'. Show that an adversary can now easily compute Alice's secret key.

(d) We have shown that Alice can choose her private key in such a way that she can later construct duplicate signatures on two messages x and x'. Does this property mean that the ECDSA is "insecure"? (Discuss.)

Symmetric-Key Cryptography

Problem 2/2025: MAC schemes.

(a) Let E denote the family of encryption functions for a block cipher where plaintext blocks, ciphertext blocks, and keys are each 128 bits in length. Let $H\colon \{0,1\}^* \to \{0,1\}^{128}$ be a hash function. Define a MAC scheme $\mathrm{MAC}_k\colon \{0,1\}^{3072} \to \{0,1\}^{128}$ by $\mathrm{MAC}_k(m) = E_k(H(m))$. Here, k is a 128-bit secret key. Is this MAC scheme secure? (Explain)

(b) Consider the following MAC scheme for authenticating fixed-length messages from $\{0,1\}^{256}$. For each i, $1 \le i \le 128$, let S_i be a fixed subset of $\{1,2,3,\ldots,255,256\}$. The key space is $\{0,1\}^{128}$. To authenticate a message m with key k, one first forms the 128-bit string $b = b_1 b_2 \cdots b_{128}$ where b_i is the sum modulo 2 of the bits of m indexed by the elements of S_i, and then AES-encrypts b with key k. The resulting ciphertext c is the tag on m.

Show how a passive adversary who is given a single valid message/tag pair can easily produce new valid message/tag pairs.

Problem 2/2020: Authenticated encryption.

Let $E\colon \mathcal{K} \times \mathcal{M} \to \mathcal{C}$ and $D\colon \mathcal{K} \times \mathcal{C} \to \mathcal{M}$ be the encryption and decryption functions for a symmetric-key encryption scheme, and let $\mathrm{MAC}\colon \mathcal{K} \times \mathcal{M} \to \mathcal{T}$ be a message authentication code. Consider the following two candidate constructions for an authenticated encryption scheme, which aims to simultaneously provide confidentiality and integrity:

(i) Split k into two keys k_1, k_2. Compute $c \leftarrow E(k_1, m)$ and $t \leftarrow \mathrm{MAC}(k_2, c)$. Return (c, t).

(ii) Split k into two keys k_1, k_2. Compute $c \leftarrow E(k_1, m)$ and $t \leftarrow \mathrm{MAC}(k_2, m)$. Return (c, t).

Assume that (E, D) constitute a secure encryption scheme, and MAC is a secure message authentication code. For each of (i) and (ii), is the candidate construction guaranteed to provide both confidentiality and integrity? If not, give a counterexample. If so, give an argument to that effect.

Problem 3/2019, 1/2017: Key whitening and DES-X.

The original DES block cipher is limited to a 56-bit key and 64-bit plaintext/ciphertext blocks. The DES-X block cipher, proposed by Ron Rivest, uses a 184-bit key (k, k_1, k_2) where $k \in \{0,1\}^{56}$ and $k_1, k_2 \in \{0,1\}^{64}$. The encryption of a plaintext $m \in \{0,1\}^{64}$ is given by

$$E(m) = \mathrm{DES}_k(m \oplus k_1) \oplus k_2,$$

where $\mathrm{DES}_k(m)$ denotes the DES-encryption of a 64-bit plaintext block m with 56-bit secret key k.

(a) Describe the decryption procedure.

(b) Suppose that the XOR with k_1 is omitted, i.e.,

$$E(m) = \mathrm{DES}_k(m) \oplus k_2,$$

where the key (k, k_2) is now 120 bits. Describe an attack that recovers the secret key using roughly 2^{56} DES operations.

(c) Suppose that the XOR with k_2 is omitted, i.e.,

$$E(m) = \mathrm{DES}_k(m \oplus k_1),$$

where the key (k, k_1) is now 120 bits. Describe an attack that recovers the secret key using roughly 2^{56} DES operations.

Problem 1/2018: Block ciphers.

Recall that DES is a block cipher with key space $K = \{0,1\}^{56}$, plaintext space $M = \{0,1\}^{64}$, and ciphertext space $C = \{0,1\}^{64}$.

(a) Let $\overline{m}$ denote the bitwise complement of a bit string m (i.e., $\overline{m} = m \oplus 11\ldots1$). By examining the description of DES, one can see that if $c = \mathrm{DES}_k(m)$ then $\overline{c} = \mathrm{DES}_{\overline{k}}(\overline{m})$. Can you use this property of DES to (slightly) improve the running time of exhaustive key search under a chosen-plaintext attack?

(b) Recall that Triple-DES has key space $K = \{0,1\}^{168}$. A plaintext $m \in \{0,1\}^{64}$ is encrypted under key $k = (k_1, k_2, k_3)$ (where $k_1, k_2, k_3 \in \{0,1\}^{56}$) as follows:

$$E_k(m) = DES_{k_3}(DES_{k_2}(DES_{k_1}(m))).$$

Describe a known-plaintext attack on Triple-DES that is significantly faster than exhaustive key search. Estimate the time and space requirements of your attack.

Problem 6/2016: Message authentication codes.

Recall the definition of CBC-MAC:

(a) Is CBC-MAC with one-block inputs existentially unforgeable under a chosen-message attack (EUF-CMA)? (b) Is CBC-MAC with variable-length inputs existentially unforgeable under a chosen-message attack (EUF-CMA)?

Problem 2/2013: Block ciphers.

Recall that the CBC block cipher mode of operation encrypts a message $m_1 m_2 \cdots m_n$ to the ciphertext $c_0 c_1 c_2 \cdots c_n$ where c_0 is chosen at random and

$$c_i = E_k(m_i \oplus c_{i-1}).$$

(a) Explain how decryption is performed with CBC.

(b) We define a new block cipher mode of operation known as UBC (Useless Block Chaining), with

$$c_i = E_k(m_i) \oplus c_{i-1}.$$

Show that UBC is, in fact, useless.

Problem 2/2011: Block ciphers.

Recall that a Feistel cipher has a round function of the following form

$$L_i^t = R_i^{t-1}, \qquad R_i^t = L_i^{t-1} \oplus f(R_i^{t-1}, K_i).$$

The plaintext is $L^0 \,||\, R^0$ and the ciphertext is $L^n \,||\, R^n$.

(a) What properties does f need to satisfy in order for encryption to be invertible? Justify.

(b) Describe the decryption algorithm.

Problem 5/2011: Pseudo-random bit generators.

Let (n, e) be an RSA public key, and let d be the corresponding RSA private key. Let $f : \mathbb{Z}_n \to \mathbb{Z}_n$ be defined by $f(x) = x^e \bmod n$. Let

$$B(x) = \begin{cases} 1, & \text{if } x^d \bmod n \text{ is odd,} \\ 0, & \text{if } x^d \bmod n \text{ is even.} \end{cases}$$

Clearly, $B(x)$ is easy to compute given x and $f^{-1}(x)$ (since $f^{-1}(x) = x^d \bmod n$). It can be shown (and you may assume) that $B(x)$ is hard to compute given only x.

(a) Define (informally) what it means for a PRBG G to pass the next bit test.

(b) Describe a cryptographically secure PRBG G which takes as seed random $x_0 \in \mathbb{Z}_n$ and uses f and B above.

(c) Prove that the generator G passes the next bit test.

Problem 1/2003: Chosen-plaintext attack on two-key Triple-DES.

Recall that DES is a block cipher with key space $K = \{0,1\}^{56}$, plaintext space $M = \{0,1\}^{64}$, and ciphertext space $C = \{0,1\}^{64}$. Encryption for *two-key Triple-DES* is defined as follows:

$$E_k(m) = \mathrm{DES}_{k_1}(\mathrm{DES}^{-1}_{k_2}(\mathrm{DES}_{k_1}(m))),$$

where $k = (k_1, k_2)$, and $k_1, k_2 \in_R \{0,1\}^{56}$ is the secret key. Design a *chosen-plaintext* attack on two-key Triple-DES that takes *roughly* 2^{56} steps. (A step is a DES encryption or decryption operation.) Provide an explanation of why your attack works, and a careful estimate of its space and time requirements. (Hint: your attack may need a *lot* of chosen plaintext/ciphertext pairs.)

Problem 1/2002: Attempts to strengthen DES against exhaustive key search attacks.

Recall that DES is a symmetric-key encryption scheme with a 56-bit key, and 64-bit plaintext and ciphertext blocks. Consider the following proposal for a new symmetric-key encryption scheme based on DES. The secret key for the new scheme is $k = (k_1, k_2)$, where $k_1 \in \{0,1\}^{56}$ and $k_2 \in \{0,1\}^{64}$ (so k is a 120-bit key). Let $m \in \{0,1\}^{64}$ be a plaintext message. Then encryption is defined as follows:

$$E_k(m) = \mathrm{DES}_{k_1}(m \oplus k_2).$$

(a) Show how this encryption scheme can be totally broken—that is, the secret key k can be recovered—by a known-plaintext attack using *roughly* 2^{56} DES encryption/decryption operations. Your attack should have little space requirements. You may assume that you have a moderate number of plaintext-ciphertext pairs $(m_i, c_i = E_k(m_i))$. Briefly justify why the number of such pairs you use is sufficient to uniquely determine the key with high probability.

(b) Is the encryption scheme with encryption function $E_k(m) = \mathrm{DES}_{k_1}(m \oplus k_2) \oplus k_2$ any more secure than DES? [Briefly justify your answer.]

(c) Is the encryption scheme with encryption function $E_k(m) = \mathrm{DES}_{k_1}(m \oplus k_2) \oplus k_3$ any more secure than DES? (Here, $k = (k_1, k_2, k_3)$ where $k_1 \in \{0,1\}^{56}$ and $k_2, k_3 \in \{0,1\}^{64}$.) [Briefly justify your answer.]

Hash Functions

Problem 1/2025: Hash functions.

Let $H\colon \{0,1\}^{\leq \ell} \to \{0,1\}^n$ be a hash function mapping binary strings of length at most ℓ, for some $\ell > n$, to binary strings of length n.

(a) Define what it means for H to be collision resistant.

(b) Define what it means for H to be second-preimage resistant.

(c) Prove one of the following statements:

(i) If H is collision resistant, then H is also second-preimage resistant.

(ii) If H is second-preimage resistant, then H is also collision resistant.

(d) Define $G\colon \{0,1\}^{\leq \ell} \to \{0,1\}^n$ by $G(x) = H(H(x))$. Prove that if H is collision resistant then G is also collision resistant.

Problem 6/2020: Hash functions.

Let p be a 256-bit prime, and let E be an elliptic curve defined over $\mathbb{Z}_p$ with $|E(\mathbb{Z}_p)| = n$ a prime. Let $P, Q \in E(\mathbb{Z}_p)$ be randomly selected points, neither of which is the point at infinity; these points are fixed and public. Define the hash function $H : [0, n-1] \times [0, n-1] \to E(\mathbb{Z}_p)$ by $H((a,b)) = aP + bQ$. That is, messages are pairs (a,b) of integers in the interval $[0, n-1]$, and the hash of such a message is the elliptic curve point $aP + bQ$.

(a) Define what it means for H to be preimage resistant.

(b) Define what it means for H to be collision resistant.

(c) Prove, under a reasonable computational assumption (which you should state), that H is collision resistant.

(d) Prove, under a reasonable computational assumption (which you should state), that H is preimage resistant.

Problem 2/2019: Hash functions.

Recall that AES is a block cipher with message space $\{0,1\}^{128}$ and key space $\{0,1\}^{128}$. Let x and y denote bitstrings of length 128. Which of the following hash functions $H_i : \{0,1\}^{256} \to \{0,1\}^{128}$, if any, are preimage resistant? No justification is needed except that a lack of preimage resistance must be justified.

(i) $H_1(x,y) = \mathrm{AES}_x(y) \oplus y$.

(ii) $H_2(x,y) = \mathrm{AES}_y(x) \oplus x$.

(iii) $H_3(x,y) = \mathrm{AES}_y(x) \oplus y$.

Problem 2/2018: Hash functions.

Let q be a prime, and let G be a (multiplicatively written) group of order q. Let g and h be randomly selected elements from $G \setminus \{1\}$. Consider the function $H_{g,h} : \mathbb{Z}_q \times \mathbb{Z}_q \to G$ defined by $H_{g,h} : (x,y) \mapsto g^x h^y$. Henceforth we will denote $H_{g,h}$ by H.

(a) Show that for any $k \in G$, there are exactly q distinct solutions $(x,y) \in \mathbb{Z}_q \times \mathbb{Z}_q$ to the equation $g^x h^y = k$.

(b) Prove that if the discrete logarithm problem in G is intractable then H is collision resistant.

(c) Is H preimage resistant?

Problem 2/2017: Hash functions.

(a) Define what it means for a hash function to be *collision resistant*.

(b) Define what it means for a hash function to be *preimage resistant*.

(c) Define what it means for a hash function to be *second-preimage resistant*.

(d) Let $G\colon \{0,1\}^{2n} \to \{0,1\}^n$ and $H\colon \{0,1\}^{2n} \to \{0,1\}^n$ be two hash functions. Define the function $F\colon \{0,1\}^{2n} \times \{0,1\}^n \to \{0,1\}^n$ by $F(x,y) = H(G(x), G(x))$. (Here, the comma "," denotes concatenation.) Prove that if G and H are collision resistant, then F is also collision resistant.

(e) Suppose that $f\colon \{0,1\}^{n+r} \to \{0,1\}^n$ is a preimage resistant function. Define the function $H\colon \{0,1\}^{2(n+r)} \to \{0,1\}^n$ as follows. Given $x \in \{0,1\}^{2(n+r)}$, write $x = x_L \,\|\, x_R$ where $x_L, x_R \in \{0,1\}^{n+r}$. (Here, $\|$ denotes concatenation.) Then define

$$H(x) = f\big(x_L \oplus x_R\big).$$

Prove that H is not second-preimage resistant.

Problem 1/2016: Hash functions.

In the triangle of the three properties of a hash function:

- collision resistant
- preimage resistant
- second-preimage resistant

enter six symbols $\in \{\Rightarrow, \Leftrightarrow\}$ to indicate which property implies the other and which does not. Prove three out of the six directions.

Problem 1/2013: Hash functions.

(a) Define what it means for a hash function to be preimage resistant.

(b) Let E be a block cipher with 80-bit keys and an 80-bit block size. Fix a publicly known 80-bit initialization vector IV. Define a hash function $H : \{0,1\}^{160} \to \{0,1\}^{80}$ as follows:

$$H(m) = E_{m_2}\big(E_{m_1}(\mathrm{IV})\big),$$

where m_1 (respectively, m_2) represents the first (respectively, last) 80 bits of m. Give (and analyze) an algorithm to find preimages for H. Your algorithm should use no more than $\approx 2^{40}$ invocations of E.

Problem 1/2011: Hash functions.

(a) Give an example of a hash function $H : \{0,1\}^* \to \{0,1\}^*$ that is collision resistant but not preimage resistant. Justify your choice!

(b) Define what it means for a pair of permutations f_0, f_1 on S to be claw-free.

(c) Let $p, q \equiv 3 \pmod 4$, and let $n = pq$. Let S denote the set of squares modulo n, coprime with n. Let $a_0, a_1 \in S$, and define the functions $f_0 : S \to S$ and $f_1 : S \to S$ by

$$f_0(x) = a_0 x^2 \pmod n, \qquad f_1(x) = a_1 x^2 \pmod n.$$

Then f_0 and f_1 are permutations on S (you do not need to show this). Show that under the assumption that factoring n is computationally infeasible, f_0 and f_1 is a claw-free pair of permutations.

(d) Using f_0 and f_1, construct a collision-free hash function

$$H : \{0,1\}^* \to S.$$

Problem 1/2007: Hash functions.

(a) Is a collision-resistant hash function necessarily preimage resistant?

(b) Let (n, e) be an RSA public key, where n is 2048 bits in length. The corresponding RSA private key is not known to anyone. Define a hash function $H : \{0,1\}^* \to [0, n-1]$ as follows: $H(m) = m^e \bmod n$, where m denotes the integer whose binary representation is m. Is H collision resistant? (Justify your answer.)

(c) Let $IV \in \{0,1\}^n$ be a fixed initialization vector, and let $f\colon \{0,1\}^{n+r} \to \{0,1\}^n$ be a compression function. Define the hash function H as follows: to hash a message x of bitlength $b < 2^r$, the message is first divided into r-bit blocks: $\bar{x} = x_1, x_2, \ldots, x_t$ (where the last block is padded with 0 bits if necessary). Define x_{t+1} to be the r-bit binary representation of b. Then $H_0 = \text{IV}$ and $H_i = f(H_{i-1}, x_i)$ for $i = 1, 2, \ldots, t$. Then $H(x)$ is defined to be H_{t+1}. Prove that if f is collision resistant, then H is also collision resistant.

Problem 1/2004: Hash functions.

(a) Define what it means for a hash function to be collision resistant. (b) Define what it means for a hash function to be second preimage resistant. (c) Let E denote the family of encryption functions for the AES block cipher where plaintext blocks, ciphertext blocks, and keys are each 128 bits in length. Define a hash function $H : \{0,1\}^{128} \to \{0,1\}^{128}$ by $H(x, y) = E_x(y)$. Here, x and y are 128-bit blocks, and $E_x(y)$ denotes the encryption of the plaintext block y using the key x. Is H collision resistant? (Justify your answer.) (d) Is the hash function in (c) second preimage resistant? (Justify your answer.)

Problem 2/2002: Hash functions.

Suppose $h_1 \colon \{0,1\}^{2m} \to \{0,1\}^m$ is a collision resistant hash function.

(a) Define $h_2 \colon \{0,1\}^{4m} \to \{0,1\}^m$ as follows:

(i) Write $x \in \{0,1\}^{4m}$ as $x = x_1 \parallel x_2$, where $x_1, x_2 \in \{0,1\}^{2m}$.

(ii) Define $h_2(x) = h_1(x_1) \parallel h_1(x_2)$.

Prove that h_2 is collision resistant.

(b) For an integer $i \geq 2$, define a hash function $h_i \colon \{0,1\}^{2^i m} \to \{0,1\}^m$ recursively from h_{i-1} as follows:

(i) Write $x \in \{0,1\}^{2^i m}$ as $x = x_1 \parallel x_2$, where $x_1, x_2 \in \{0,1\}^{2^{i-1} m}$.

(ii) Define $h_i(x) = h_{i-1}(x_1) \parallel h_{i-1}(x_2)$.

Prove that h_i is collision resistant.

(c) Let γ be a quadratic residue modulo p. Show that the two square roots of γ modulo p are $\pm\gamma^{(p+1)/4}$.

(d) Show that $L_1(\beta)$ can be efficiently computed given p, α, β. Here, $L_1(\beta)$ denotes the least significant bit of the discrete logarithm a of $\beta = \alpha^a \pmod{p}$ (i.e., $L_1(\beta) = 0$ if a is even, and $L_1(\beta) = 1$ if a is odd). Let $L_2(\beta)$ denote the second least significant bit of a. That is, $L_2(\beta) = 0$ if $a \equiv 0$ or $1 \pmod{4}$, and $L_2(\beta) = 1$ if $a \equiv 2$ or $3 \pmod{4}$.

RSA Cryptosystem

Problem 4/2020, 4/2017: Integer factorization.

(a) Describe the *random squares method* for factoring a number n that is not a prime or a prime power. You are not expected to analyze the running time of the algorithm. (Note: In Stinson's book, the algorithm is called "Dixon's random squares algorithm". In Koblitnz's book, the algorithm is called "Factor base algorithm".)

(b) Explain the trade-off that dictates the optimal size of the factor base.

Problem 4/2019: Cryptanalysis of RSA.

Suppose that Alice and Bob agree to generate and share a common RSA modulus $n = pq$ for use in RSA encryption. Alice chooses an encryption exponent e_a and the corresponding decryption exponent d_a, and Bob chooses an encryption exponent e_b and the corresponding decryption exponent d_b. Suppose furthermore $\gcd(e_a, e_b) = 1$. Show that an adversary who acquires a pair of ciphertexts $c_a \equiv m^{e_a} \pmod{n}$ and $c_b \equiv m^{e_b} \pmod{n}$ corresponding to a single message m encrypted to Alice and Bob respectively can recover the plaintext message m efficiently.

Problem 5/2018: RSA.

(a) Suppose that Alice's RSA public key is $(n = 143, e = 7)$. Determine her private key d.

(b) Let (n, e) be an RSA public key, where $n = pq$, and e is an integer with $1 < e < \phi(n)$ and $\gcd(e, \phi(n)) = 1$. It is known that the number of plaintexts $m \in [0, n-1]$ satisfying $m^e \equiv m \pmod{n}$ is

$$[1 + \gcd(e-1, p-1)] \cdot [1 + \gcd(e-1, q-1)].$$

Such a plaintext message m is called an *unconcealed message* since its RSA cyphertext is equal to m itself. Prove that there is at least one value of e, $1 < e < \phi(n)$, $\gcd(e, \phi(n)) = 1$, such that $m^e \equiv m \pmod{n}$ for all $m \in [0, n-1]$.

Problem 4/2016: RSA.

Suppose that textbook RSA is used to encrypt a random 56-bit DES key k without padding; that is, the value of k as an integer is used as an RSA plaintext. Given the corresponding RSA ciphertext, give a (classical) algorithm that, with high probability, recovers the key k in substantially fewer than 2^{56} operations. Hint: Use the fact that a random integer 56-bit integer factors into a product of two integers less than 2^{29} with high probability.

Problem 5/2013: RSA.

Alice has a corrupted copy of Bob's RSA public key (n, e), in which one bit of the public exponent e is wrong. Alice encrypts a message m under the textbook RSA scheme using this corrupted public key, and sends the resulting ciphertext c_1 to Bob. Later, Alice obtains a correct copy of Bob's RSA public key, and sends an encryption c_2 of the same message m under textbook RSA using the correct key. An adversary, who knows Bob's public key, obtains both c_1 and c_2. Show how the adversary can obtain m.

Problem 3/2003: Partial key-exposure in RSA.

This exercise shows that if the encryption exponent in RSA is $e = 3$, then the left half of the bits of d can be very easily computed. (More precisely, the possible values for the left half of the bits can be narrowed to one or two.)

Let $n = pq$ where p and q are primes with $5 \leq p < q < 2p$. Let integers e and d satisfy $1 < e, d < \phi(n)$ and $ed \equiv 1 \pmod{\phi(n)}$.

(a) Prove that there exists an integer k satisfying $ed - k\phi(n) = 1$ and $1 \leq k < e$.

(b) Let $\hat{d} = \lfloor \frac{kn+1}{e} \rfloor$. Prove that $|\hat{d} - d| < 3\sqrt{n}$.

(c) Prove that if $e = 3$ then $k = 2$.

Problem 4/2002: Speeding up RSA decryption.

Let (n, e) be Alice's RSA public key, and let d be her corresponding private key. Recall that the RSA encryption operation is $c = m^e \bmod n$, while the RSA decryption operation is $m = c^d \bmod n$. One way to speed up RSA decryption is to precompute $d_p = d \bmod (p-1)$ and $d_q = d \bmod (q-1)$. Then decryption of a ciphertext c can be performed by computing $m_p = c^{d_p} \bmod p$ and $m_q = c^{d_q} \bmod q$, and then finding m, $0 \leq m \leq n-1$, such that $m \equiv m_p \pmod{p}$ and $m \equiv m_q \pmod{q}$.

(a) Describe a procedure (i.e., a formula) that Alice can use to compute m efficiently, given m_p and m_q.

(b) Prove that m is the correct decryption of c. That is, prove that $m \equiv c^d \pmod{n}$.

(c) Briefly justify the assertion that this method of decryption can speed up RSA decryption by approximately 75

(d) Devise an algorithm which, on input n and e, factors n in $O(\min(d_p, d_q))$ steps. (A "step" is any operation whose running time is polynomial in $\log n$.) State any assumptions you may make. [This exercise shows that Alice should not try to speed up decryption by selecting d so that d_p and d_q are too small.]

Part II.

Solutions grouped by topic

Elementary Number Theory

Problem 3/2025, 3/2020: Primality testing (topic).

Recall that if n is a composite integer, and $a \in [1, n-1]$ satisfies $a^{n-1} \equiv 1 \pmod{n}$, then a is called a Fermat liar for n.

Now, let $n = pq$, where p and q are distinct odd primes, let $a \in \mathbb{Z}_n^\times$, and let $d = \gcd(p-1, q-1)$.

(a) Prove that a is a Fermat liar for n if and only if $a^d \equiv 1 \pmod{n}$.

(Hint: Note that $n - 1 = pq - 1 = (p-1)q + (q-1)$.)

(b) Suppose now that $q = 2p + 1$. How many $a \in \mathbb{Z}_n^*$ are Fermat liars for n? List them all (in terms of p).

Solution

(a) On the one hand, assume $a^d \equiv 1 \pmod{n}$. Since $d \mid p-1$ and $d \mid q-1$, we have $d \mid (p-1)q + (q-1) = n - 1$. Therefore, $a^{n-1} \equiv (a^d)^{(n-1)/d} \equiv 1^{(n-1)/d} \equiv 1 \pmod{n}$, so a is a Fermat liar for n.

On the other hand, assume a is a Fermat liar for n, i.e., $a^{n-1} \equiv 1 \pmod{n}$. Therefore, $a^{n-1} \equiv 1 \pmod{p}$. By Fermat's little theorem, $a^{p-1} \equiv 1 \pmod{p}$. Therefore,

$$a^{q-1} \equiv a^{(n-1)-(p-1)q} \equiv 1 \cdot 1^{-q} \equiv 1 \pmod{p}.$$

Since $d = \gcd(p-1, q-1)$, by the extended Euclidean algorithm, there exist integers x and y such that $d = x(p-1) + y(q-1)$. Therefore,

$$a^d \equiv a^{x(p-1)+y(q-1)} \equiv (a^{p-1})^x (a^{q-1})^y \equiv 1^x 1^y \equiv 1 \pmod{p}.$$

Similarly, we can show that $a^d \equiv 1 \pmod{q}$. By the Chinese remainder theorem, we conclude that $a^d \equiv 1 \pmod{n}$.

(b) Since $q = 2p + 1$, we have $d = \gcd(p - 1, q - 1) = \gcd(p - 1, 2p) = \gcd(p - 1, 2) = 2$. Therefore, a is a Fermat liar for n if and only if $a^2 \equiv 1 \pmod{n}$. This is equivalent to $a^2 \equiv 1 \pmod{p}$ and $a^2 \equiv 1 \pmod{q}$, which is equivalent to $a \equiv \pm 1 \pmod{p}$ and $a \equiv \pm 1 \pmod{q}$.

Each of the four combinations of signs gives a unique solution modulo n by the Chinese remainder theorem:

- $a \equiv 1 \pmod{p}$ and $a \equiv 1 \pmod{q}$ gives
$$a \equiv 1 \pmod{n}.$$
- $a \equiv 1 \pmod{p}$ and $a \equiv -1 \pmod{q}$ gives
$$a \equiv 4p + 1 = 2q - 1 \pmod{n}.$$
- $a \equiv -1 \pmod{p}$ and $a \equiv 1 \pmod{q}$ gives
$$a \equiv -4p - 1 = -2q + 1 \pmod{n}.$$
- $a \equiv -1 \pmod{p}$ and $a \equiv -1 \pmod{q}$ gives
$$a \equiv -1 \pmod{n}.$$

Problem 1/2019: Elementary number theory (topic).

Let $n = pq$ be an RSA modulus. Prove the following:

$$\prod_{x \in \mathbb{Z}_n^*} x \equiv 1 \pmod{n}.$$

Solution

Consider the involution of $\mathbb{Z}_n^*$ defined by mapping each element x to its inverse x^{-1}. If x is not a fixed point of this involution, then it pairs with x^{-1} such that their product is congruent to 1 modulo n. Therefore, the left-hand side of the desired congruence can be expressed as the product of all fixed points of the involution.

The fixed points of this involution are the elements x such that $x \equiv x^{-1} \pmod{n}$, or equivalently, $x^2 - 1 \equiv 0 \pmod{n}$. To find these fixed points, we can use the Chinese Remainder Theorem. We need to simultaneously solve $x^2 \equiv 1 \pmod{p}$ and $x^2 \equiv 1 \pmod{q}$. Each of these congruences has exactly two solutions: $x \equiv 1 \pmod{p}$ and $x \equiv -1 \pmod{p}$ for the first, and $x \equiv 1 \pmod{q}$ and $x \equiv -1 \pmod{q}$ for the second.

By the Chinese Remainder Theorem, there are four solutions r_1, r_2, r_3, r_4 to the system of congruences, corresponding to the combinations of the solutions modulo p and q:

- $x \equiv 1 \pmod{p}$ and $x \equiv 1 \pmod{q}$ gives $x \equiv r_1 \equiv 1 \pmod{n}$.
- $x \equiv 1 \pmod{p}$ and $x \equiv -1 \pmod{q}$ gives $x \equiv r_2 \pmod{n}$.
- $x \equiv -1 \pmod{p}$ and $x \equiv 1 \pmod{q}$ gives $x \equiv r_3 \pmod{n}$.
- $x \equiv -1 \pmod{p}$ and $x \equiv -1 \pmod{q}$ gives $x \equiv r_4 \equiv -1 \pmod{n}$.

The four fixed points are therefore $r_1 \equiv 1$, r_2, r_3, and $r_4 \equiv -1$ modulo n. Therefore, the product of all elements in $\mathbb{Z}_n^*$ is congruent to $r_1 \cdot r_2 \cdot r_3 \cdot r_4 \pmod{n}$.

Modulo p, such product is congruent to $(1) \cdot (1) \cdot (-1) \cdot (-1) \equiv 1 \pmod{p}$. Similarly, modulo q, the product is congruent to $(1) \cdot (-1) \cdot (1) \cdot (-1) \equiv 1 \pmod{q}$. Finally, by the Chinese Remainder Theorem, we conclude that

$$\prod_{x \in \mathbb{Z}_n^*} x \equiv r_1 \cdot r_2 \cdot r_3 \cdot r_4 \equiv 1 \pmod{n}.$$

Problem 3/2018: Elementary number theory (topic).

(a) Let $n \geq 3$ be an integer. Suppose that there exists an integer a such that $a^{n-1} \equiv 1 \pmod{n}$ and $a^{(n-1)/q} \not\equiv 1 \pmod{n}$ for all prime divisors q of $n-1$. Prove that n is prime.

(b) The *Fermat numbers* are $F_k = 2^{2^k} + 1$ for $k \geq 1$. Prove that for $k \geq 2$, F_k is prime if and only if $5^{(F_k-1)/2} \equiv -1 \pmod{F_k}$. (It may help to remember Euler's Theorem: If p is an odd prime, then $(\frac{a}{p}) \equiv a^{(p-1)/2} \pmod{p}$.)

Solution

(a) Since $a^{n-1} \equiv 1 \pmod{n}$, $a \in \mathbb{Z}_n^\times$ and its order $d = \mathrm{ord}_n(a)$ divides $n-1$. If $d < n-1$, then $(n-1)/d$ has a prime factor q, which implies $d \mid (n-1)/q$ and thus $a^{(n-1)/q} \equiv 1 \pmod{n}$, contradicting the hypothesis. Therefore, $\mathrm{ord}_n(a) = n-1$.

Since $\mathrm{ord}_n(a) = n-1$, the set $\{1, a, \ldots, a^{n-2}\}$ consists of $n-1$ distinct elements in $\mathbb{Z}_n^\times$. We therefore have $\{1, a, \ldots, a^{n-2}\} \subseteq \mathbb{Z}_n^\times \subseteq \{1, \ldots, n-1\}$. Since the outer sets both have cardinality $n-1$, it follows that $\mathbb{Z}_n^\times = \{1, \ldots, n-1\}$. Hence every integer in $\{1, \ldots, n-1\}$ is coprime to n, which proves that n is prime.

(b) On the one hand, assume F_k is prime. By Euler's criterion, $5^{(F_k-1)/2} \equiv \left(\frac{5}{F_k}\right) \pmod{F_k}$. By quadratic reciprocity, $\left(\frac{5}{F_k}\right) = \left(\frac{F_k}{5}\right)$ since $5 \equiv 1$ and $F_k \equiv 1 \pmod{4}$. For $k \geq 2$,

$$F_k = 2^{2^k} + 1 = (2^4)^{2^{k-2}} + 1 \equiv 1^{2^{k-2}} + 1 \equiv 2 \pmod{5}.$$

Since 2 is a quadratic non-residue modulo 5, $\left(\frac{2}{5}\right) = -1$, so

$$5^{(F_k-1)/2} \equiv \left(\frac{F_k}{5}\right) = \left(\frac{2}{5}\right) = -1 \pmod{F_k}.$$

On the other hand, assume $5^{(F_k-1)/2} \equiv -1 \pmod{F_k}$. Squaring yields $5^{F_k-1} \equiv (-1)^2 \equiv 1 \pmod{F_k}$. Since $F_k - 1 = 2^{2^k}$, its only prime divisor is $q = 2$. For this only prime divisor, $5^{(F_k-1)/2} \equiv -1 \not\equiv 1 \pmod{F_k}$ (as $F_k > 2$). Thus $a = 5$ and $n = F_k$ satisfy the conditions of part (a), so F_k is prime.

Problem 3/2017: Elementary number theory (topic).

Note: Parts (a) and (b) are unrelated.

(a) Let p be a prime, $n \in \mathbb{N}$, and $q = p^n$. Prove that the finite field $\mathbb{F}_q$ has $q-2$ generators if and only if $q-1$ is a Mersenne prime.

(b) Let $m \geq 3$ be an integer. Prove that if a is a quadratic residue modulo m, and $ab \equiv 1 \pmod{m}$, then b is also a quadratic residue. Now let p be a prime of the form $p = 4k+3$. Prove that the product of all the quadratic residues modulo p is congruent to 1.

Incomplete Solution

(a)

(b) Since a is a quadratic residue modulo m, there exists an integer x such that $x^2 \equiv a \pmod{m}$. Then, $x(xb) = 1 \pmod{m}$, so xb is the multiplicative inverse of x modulo m. Therefore, $b \equiv (xb)^2 \pmod{m}$, so b is also a quadratic residue modulo m.

Now let $p = 4k+3$. The nonzero quadratic residues modulo p form a set closed under inversion, by the first part. Pair each residue x with its inverse x^{-1}. Every such pair contributes 1 to the product, except possibly the fixed points of the pairing. These satisfy $x = x^{-1}$, hence $x^2 \equiv 1 \pmod{p}$, so $x \equiv \pm 1 \pmod{p}$. Since -1 is a quadratic non-residue when $p \equiv 3 \pmod{4}$, only 1 belongs to the set of quadratic residues. Thus

$$\prod_{x \text{ is a quadratic residue modulo } p} x \equiv \prod_{x \text{ is a fixed point}} x \equiv 1 \pmod{p}.$$

Problem 2/2016, 2/2003: Fermat primes and generators (topic).

Let $p = 2^m + 1$ be a prime, with $m \geq 1$.

(a) Prove that m is a power of 2.

(b) Prove that any quadratic nonresidue modulo p is a generator of $\mathbb{F}_p^\times$.

(c) Prove that 5 is a generator of $\mathbb{F}_p^\times$, except in the case $p = 5$.

(d) Prove that 7 is a generator of $\mathbb{F}_p^\times$, except in the case $p = 3$.

Solution

(a) Assume $k \geq 3$ is odd and $k \mid m$. Using that $a - b \mid a^k - b^k$ for any $a, b \in \mathbb{Z}$, substituting $a = 2^{m/k}$ and $b = -1$ we get $2^{m/k} + 1 \mid 2^m - (-1)^k = 2^m + 1$. Since $1 < 2^{m/k} + 1 < 2^m + 1$, we conclude that $2^{m/k} + 1$ is a nontrivial divisor of p, which is impossible since p is prime.

Therefore, m has no odd divisors greater than 1, which implies that m is a power of 2.

(b) Let g be a primitive root modulo $p = 2^m + 1$. On the one hand, the set of quadratic non-residues modulo p is

$$\overline{\mathrm{QR}}_p = \{g^k : k \text{ odd}\}.$$

On the other hand, the set of generators of $\mathbb{F}_p^\times$ is

$$\mathrm{Gen}_p = \{g^k : \gcd(k, p-1) = 1\}.$$

Since $p - 1 = 2^m$, we have $\gcd(k, p-1) = 1$ if and only if k is odd. Therefore, $\overline{\mathrm{QR}}_p = \mathrm{Gen}_p$, which proves that any quadratic nonresidue modulo p is a generator of $\mathbb{F}_p^\times$.

(c) By part (a), write $m = 2^k$ for some $k \geq 0$. For $k = 0$, we have $p = 3$ and $5 \equiv 2 \pmod 3$ is a generator of $\mathbb{F}_3^\times$. For $k = 1$, we have $p = 5$, which is excluded in the statement. Henceforth, assume $k \geq 2$.

Note that $p = 2^{2^k} + 1 = (2^4)^{2^{k-2}} + 1 = 16^{2^{k-2}} + 1 \equiv 1^{2^{k-2}} + 1 \equiv 2 \pmod 5$. Moreover, the set of quadratic residues modulo 5 is $\{1, 4\}$, and hence the set of quadratic nonresidues modulo 5 is $\{2, 3\}$.

Therefore, using quadratic reciprocity, since $5 \equiv 1 \pmod 4$, we have

$$\left(\frac{5}{p}\right) = \left(\frac{p}{5}\right) = \left(\frac{2}{5}\right) = -1,$$

which shows that 5 is a quadratic nonresidue modulo p. By part (b), we conclude that 5 is a generator of $\mathbb{F}_p^\times$.

(d) By part (a), write $m = 2^k$ for some $k \geq 0$. For $k = 0$, we have $p = 3$, which is excluded in the statement. Henceforth, assume $k \geq 1$.

Note that $2^3 \equiv 1 \pmod 7$, therefore powers of 2 modulo 7 are periodic with period 3. Moreover, the set of quadratic residues modulo 7 is $\{1, 2, 4\}$, and hence the set of quadratic nonresidues modulo 7 is $\{3, 5, 6\}$.

If k is even, say $k = 2\ell$, then $2^k \equiv (2^2)^\ell \equiv 1^\ell = 1 \pmod 3$, and hence $p = 2^{2^k} + 1 \equiv 2^1 + 1 \equiv 3 \pmod 7$. If k is odd, say $k = 2\ell + 1$, then $2^k \equiv 2^{2\ell+1} \equiv 2 \cdot (2^2)^\ell \equiv 2 \cdot 1^\ell = 2 \pmod 3$, and hence $p = 2^{2^k} + 1 \equiv 2^2 + 1 \equiv 5 \pmod 7$. Therefore, $p \equiv 3$ or $5 \pmod 7$. In either case, p is a quadratic nonresidue modulo 7.

By quadratic reciprocity, since $p \equiv 1 \pmod 4$,

$$\left(\frac{7}{p}\right) = \left(\frac{p}{7}\right) = -1$$

which shows that 7 is a quadratic nonresidue modulo p. By part (b), we conclude that 7 is a generator of $\mathbb{F}_p^\times$.

Problem 3/2016: Elementary number theory (topic).

Let n be an RSA modulus. Does n always, sometimes, or never have a primitive root? (Recall that a primitive root modulo n is an element of order $\phi(n)$ in the multiplicative group of units $\mathbb{Z}_n^*$.)

Solution

Let $n = pq$ be an RSA modulus, where p and q are distinct odd primes. The multiplicative group of units $\mathbb{Z}_n^*$ has order $\phi(n) = (p-1)(q-1)$. We prove that $\mathbb{Z}_n^*$ has no primitive roots by showing that the order of any element $g \in \mathbb{Z}_n^*$ is strictly less than $\phi(n)$.

Let $g \in \mathbb{Z}_n^*$. Let

$$t = \frac{\phi(n)}{2} = \frac{(p-1)(q-1)}{2} = (p-1)\frac{(q-1)}{2} = (q-1)\frac{(p-1)}{2},$$

which is an integer and a common multiple of both $p-1$ and $q-1$, since p and q are odd. By Fermat's little theorem, we have $g^{p-1} \equiv 1 \pmod{p}$ and $g^{q-1} \equiv 1 \pmod{q}$. Since t is a common multiple of both $p-1$ and $q-1$, we have $g^t \equiv 1 \pmod{p}$ and $g^t \equiv 1 \pmod{q}$. By the Chinese remainder theorem, we have $g^t \equiv 1 \pmod{n}$. Therefore, the order of g in $\mathbb{Z}_n^*$ divides t, which is strictly less than $\phi(n)$. Thus, g is not a primitive root modulo n. Since g was arbitrary, we conclude that $\mathbb{Z}_n^*$ has no primitive roots.

Problem 3/2013: Elementary number theory (topic).

Let p denote an odd prime.

(a) Let q be a generator (a.k.a. a primitive root) modulo p. Show that q is not a quadratic residue modulo p.

(b) Suppose $y \equiv g^x \pmod{p}$ for some integer $0 < x < p-1$. Show how one can efficiently find the least significant bit of the binary expansion of x.

Incomplete Solution

(a) By Euler's criterion, q is a quadratic residue modulo p if and only if $q^{(p-1)/2} \equiv 1 \pmod{p}$. Since q is a generator modulo p, it has order $p-1$. Therefore, $q^{(p-1)/2} \not\equiv 1 \pmod{p}$, and hence q is not a quadratic residue modulo p.

(b)

Problem 2/2007: Elementary number theory (topic).

(a) Let p be an odd prime. Prove that a quadratic residue modulo p can never be a generator of $\mathbb{F}_p^*$.

(b) Let p be an odd prime. Prove that -3 is a quadratic residue modulo p if and only if $p \equiv 1 \pmod 3$.

(c) Let $p > 3$ be a Mersenne prime. Prove that 3 is a quadratic nonresidue modulo p.

Solution

(a) Let g be a quadratic residue modulo p. By Euler's criterion, we have $g^{(p-1)/2} \equiv 1 \pmod p$. Therefore, the order of g in $\mathbb{F}_p^*$ divides $(p-1)/2$, which is strictly less than $p-1$. Hence, g cannot be a generator of $\mathbb{F}_p^*$.

(b) Recall the law of quadratic reciprocity, which states that for distinct odd primes p and q:

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}.$$

Also recall that -1 is a quadratic residue modulo p if and only if $p \equiv 1 \pmod 4$; in other words,

$$\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}.$$

Using these two facts with $q = 3$, we have

$$\left(\frac{-3}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{3}{p}\right) = (-1)^{\frac{p-1}{2}}\left(\frac{p}{3}\right)(-1)^{\frac{p-1}{2}\cdot 1} = \left(\frac{p}{3}\right) = \begin{cases} 1 & \text{if } p \equiv 1 \pmod 3, \\ -1 & \text{if } p \equiv 2 \pmod 3. \end{cases}$$

Therefore, -3 is a quadratic residue modulo p if and only if $p \equiv 1 \pmod 3$.

(c) Let $p = 2^n - 1$ be a Mersenne prime with $n > 2$. If n is even, say $n = 2k$, then $p = 2^{2k} - 1 = (2^k - 1)(2^k + 1)$ is composite, which is a contradiction. Therefore, n must be odd, say $n = 2k + 1$. Then we have

$$p = 2^{2k+1} - 1 = 4^k \cdot 2 - 1 \equiv 1^k \cdot 2 - 1 \equiv 1 \pmod 3.$$

Since $n > 2$, we have $p \equiv -1 \pmod 4$, so $\frac{p-1}{2}$ is odd. Therefore, by quadratic reciprocity, we have

$$\left(\frac{3}{p}\right) = \left(\frac{p}{3}\right)(-1)^{\frac{p-1}{2}\cdot\frac{3-1}{2}} = \left(\frac{1}{3}\right)(-1) = -1,$$

and consequently, 3 is a quadratic nonresidue modulo p.

Problem 3/2002: Carmichael numbers (topic).

Assume throughout this question that n is square-free (i.e., n is not divisible by the square of a prime). Then $n = p_1 \cdots p_\ell$, where $p_1, \ldots, p_\ell$ are distinct primes. Such an integer n is a Carmichael number if $a^{n-1} \equiv 1 \pmod{n}$ for all integers a that are relatively prime to n. You may use the following facts in your solutions: (i) $a^{n-1} \equiv 1 \pmod{p_i}$ for all i implies $a^{n-1} \equiv 1 \pmod{n}$; (ii) for every prime p, there exists a primitive element mod p.

(a) Suppose that n is a Carmichael number. Prove that $p_i - 1$ divides $n - 1$ for all $i, 1 \leq i \leq \ell$.

(b) Suppose that $p_i - 1$ divides $n - 1$ for all $i, 1 \leq i \leq \ell$. Then prove that n is a Carmichael number.

(c) Prove that 561 is a Carmichael number.

Solution

(a) Fix i, and consider the prime p_i. Let g be a primitive element mod p_i, that is, the order of g in $\mathbb{Z}_{p_i}^*$ is $p_i - 1$. By the Chinese Remainder Theorem, there exists an integer a such that $a \equiv g \pmod{p_i}$ and $a \equiv 1 \pmod{p_j}$ for all $j \neq i$. In particular, a is relatively prime to n. Since n is a Carmichael number, we have $a^{n-1} \equiv 1 \pmod{n}$, and $g^{n-1} \equiv a^{n-1} \equiv 1 \pmod{p_i}$. Therefore, the order of g divides $n - 1$, and hence $p_i - 1$ divides $n - 1$.

(b) Let a be an integer that is relatively prime to n. Then a is relatively prime to each p_i, and hence $a^{p_i - 1} \equiv 1 \pmod{p_i}$ by Fermat's little theorem. Since $p_i - 1$ divides $n - 1$, we have $a^{n-1} \equiv 1 \pmod{p_i}$ for all i. By the Chinese Remainder Theorem, we conclude that $a^{n-1} \equiv 1 \pmod{n}$. Since a was arbitrary, we conclude that n is a Carmichael number.

(c) The prime factorization of 561 is $3 \cdot 11 \cdot 17$. We have $3 - 1 = 2$, $11 - 1 = 10$, and $17 - 1 = 16$, all of which divide $560 = 561 - 1$. Thus, we conclude that $p_i - 1$ divides $n - 1$ for all i. By part (b), we conclude that 561 is a Carmichael number.

Part III.

Problems grouped by year

Problems from the 2025 exam

Problem 1/2025: Hash functions (topic).

Let $H\colon \{0,1\}^{\leq \ell} \to \{0,1\}^n$ be a hash function mapping binary strings of length at most ℓ, for some $\ell > n$, to binary strings of length n.

(a) Define what it means for H to be collision resistant.

(b) Define what it means for H to be second-preimage resistant.

(c) Prove one of the following statements:

(i) If H is collision resistant, then H is also second-preimage resistant.

(ii) If H is second-preimage resistant, then H is also collision resistant.

(d) Define $G\colon \{0,1\}^{\leq \ell} \to \{0,1\}^n$ by $G(x) = H(H(x))$. Prove that if H is collision resistant then G is also collision resistant.

Problem 2/2025: MAC schemes (topic).

(a) Let E denote the family of encryption functions for a block cipher where plaintext blocks, ciphertext blocks, and keys are each 128 bits in length. Let $H\colon \{0,1\}^* \to \{0,1\}^{128}$ be a hash function. Define a MAC scheme $\mathrm{MAC}_k\colon \{0,1\}^{3072} \to \{0,1\}^{128}$ by $\mathrm{MAC}_k(m) = E_k(H(m))$. Here, k is a 128-bit secret key. Is this MAC scheme secure? (Explain)

(b) Consider the following MAC scheme for authenticating fixed-length messages from $\{0,1\}^{256}$. For each i, $1 \leq i \leq 128$, let S_i be a fixed subset of $\{1,2,3,\ldots,255,256\}$. The key space is $\{0,1\}^{128}$. To authenticate a message m with key k, one first forms the 128-bit string $b = b_1 b_2 \cdots b_{128}$ where b_i is the sum modulo 2 of the bits of m indexed by the elements of S_i, and then AES-encrypts b with key k. The resulting ciphertext c is the tag on m.

Show how a passive adversary who is given a single valid message/tag pair can easily produce new valid message/tag pairs.

Problem 3/2025: Primality testing (solution, topic).

Recall that if n is a composite integer, and $a \in [1, n-1]$ satisfies $a^{n-1} \equiv 1 \pmod{n}$, then a is called a Fermat liar for n.

Now, let $n = pq$, where p and q are distinct odd primes, let $a \in \mathbb{Z}_n^\times$, and let $d = \gcd(p-1, q-1)$.

(a) Prove that a is a Fermat liar for n if and only if $a^d \equiv 1 \pmod{n}$.

(Hint: Note that $n - 1 = pq - 1 = (p-1)q + (q-1)$.)

(b) Suppose now that $q = 2p + 1$. How many $a \in \mathbb{Z}_n^*$ are Fermat liars for n? List them all (in terms of p).

Problem 4/2025: RSA signatures (topic).

Recall that in the Full-Domain Hash (FDH) RSA signature scheme, an entity (Alice) with public key (n, e) and private key d generates a signature s on a message m by computing $s = H(m)^d \bmod n$. Here, $H\colon \{0,1\}^* \to [0, n-1]$ is a hash function.

(a) Describe how someone can verify Alice's signature s on a message m.

(b) Prove that if finding e-th roots modulo n is intractable, and if H is a random function, then RSA-FDH is secure (i.e., existentially unforgeable by an adversary who can mount an adaptive chosen-message attack). Your proof can be informal, but should contain all the essential ideas of a formal proof.

Problem 5/2025: Discrete logarithms (topic).

Let p be a prime, let q be a prime divisor of $p-1$, and let g be an element of order q in $\mathbb{Z}_p^*$. The discrete logarithm problem is the following: given p, q, g and $h \in_R \langle g \rangle$, determine the integer $x \in [0, n-1]$ such that $h = g^x \pmod{p}$. Here, $\langle g \rangle$ denotes the subgroup of $\mathbb{Z}_p^*$ generated by g.

Describe and analyze an algorithm for solving the discrete logarithm problem that has running time $O(\sqrt{q})$ $\mathbb{Z}_p$ operations. There is no limit on how much space your algorithm can use.

Problem 6/2025: Elliptic curves (topic).

Let p be an odd prime satisfying $p \equiv 2 \pmod 3$. Consider the elliptic curve $E\colon Y^2 = X^3 + b$ defined over $\mathbb{F}_p$ ($b \neq 0$).

(a) Prove that the mapping $x \mapsto x^3$ is a bijection on $\mathbb{F}_p$.

(b) Prove that the number of points in $E(\mathbb{F}_p)$ is $p + 1$.

(c) Let $R = (x, y)$ be a point in $E(\mathbb{F}_p)$. Given y, explain how to compute x efficiently.

Problem 7/2025: Elliptic curve Diffie-Hellman key agreement (topic).

Let E be an elliptic curve defined over $\mathbb{Z}_p$, where p is a prime. Let $n = \#E(\mathbb{Z}_p)$, and suppose that n is prime. Let $P \in E(\mathbb{Z}_p)$ with $P \neq \infty$.

(a) Recall that in the unauthenticated elliptic curve Diffie-Hellman key agreement protocol (ECDH), Alice selects $x \in_R [1, n-1]$ and sends $X = xP$ to Bob. Similarly, Bob selects $y \in_R [1, n-1]$ and sends $Y = yP$ to Alice. Their shared secret key is $k = H(K)$ where $K = xyP$ and H is a hash function. Alice and Bob subsequently use k to encrypt and decrypt messages using AES.

Describe the malicious-intruder-in-the-middle attack (also known as the man-in-the-middle attack) on ECDH.

(b) Describe a variant of authenticated ECDH, and give an informal argument for its security against active attacks.

Problems from the 2020 exam

Problem 1/2020: Shannon's theory (topic).

(a) Define what it means for a symmetric-key encryption scheme to have perfect secrecy.

(b) Suppose that a symmetric-key encryption scheme with encryption function $E\colon \mathcal{K} \times \mathcal{M} \to \mathcal{C}$ satisfies $|\mathcal{K}| = |\mathcal{C}| = |\mathcal{M}|$. Prove that, if the encryption scheme has perfect secrecy, then every key is used with equal probability $1/|\mathcal{K}|$, and for every message $x \in \mathcal{M}$ and ciphertext $y \in \mathcal{C}$, there is a unique key k such that $E(k, x) = y$.

Problem 2/2020: Authenticated encryption (topic).

Let $E\colon \mathcal{K} \times \mathcal{M} \to \mathcal{C}$ and $D\colon \mathcal{K} \times \mathcal{C} \to \mathcal{M}$ be the encryption and decryption functions for a symmetric-key encryption scheme, and let $\mathrm{MAC}\colon \mathcal{K} \times \mathcal{M} \to \mathcal{T}$ be a message authentication code. Consider the following two candidate constructions for an authenticated encryption scheme, which aims to simultaneously provide confidentiality and integrity:

(i) Split k into two keys k_1, k_2. Compute $c \leftarrow E(k_1, m)$ and $t \leftarrow \mathrm{MAC}(k_2, c)$. Return (c, t).

(ii) Split k into two keys k_1, k_2. Compute $c \leftarrow E(k_1, m)$ and $t \leftarrow \mathrm{MAC}(k_2, m)$. Return (c, t).

Assume that (E, D) constitute a secure encryption scheme, and MAC is a secure message authentication code. For each of (i) and (ii), is the candidate construction guaranteed to provide both confidentiality and integrity? If not, give a counterexample. If so, give an argument to that effect.

Problem 3/2020: Primality testing (solution, topic).

Recall that if n is a composite integer, and $a \in [1, n-1]$ satisfies $a^{n-1} \equiv 1 \pmod{n}$, then a is called a *Fermat liar* for n.

Now, let $n = pq$, where p and q are distinct odd primes, let $a \in \mathbb{Z}_n^*$, and let $d = \gcd(p-1, q-1)$. ($\mathbb{Z}_n^*$ is the set of numbers in $[1, n-1]$ that are invertible modulo n.)

(a) Prove that a is a Fermat liar for n if and only if $a^d \equiv 1 \pmod{n}$. (Hint: Note that $n-1 = pq-1 = (p-1)q + (q-1)$.)

(b) Suppose now that $q = 2p+1$. How many $a \in \mathbb{Z}_n^*$ are Fermat liars for n? List them all (in terms of p).

Problem 4/2020: Integer factorization (topic).

(a) Describe the *random squares method* for factoring a number n that is not a prime or a prime power. You are not expected to analyze the running time of the algorithm. (Note: In Stinson's book, the algorithm is called "Dixon's random squares algorithm". In Koblitnz's book, the algorithm is called "Factor base algorithm".)

(b) Explain the trade-off that dictates the optimal size of the factor base.

Problem 5/2020: Discrete algorithm and Diffie-Hellman problems (topic).

Let G be a multiplicatively-written group of prime order n generated by g. The Diffie–Hellman Problem (DHP) is the following: given $g^a, g^b \in G$, compute g^{ab}.

(a) The problem SQUARE is the following: given $g^x \in G$, compute g^{x^2}. Prove that DHP $\leq_P$ SQUARE. (Recall that the notation $A \leq_P B$ means that problem A polynomial-time reduces to problem B.)

(b) The problem CUBE is the following: given $g^x \in G$, compute g^{x^3}. Prove that CUBE $\leq_P$ SQUARE.

(c) Prove that DHP $\leq_P$ CUBE. (Hint: Recall that $(x+1)^3 = x^3 + 3x^2 + 3x + 1$.)

Problem 6/2020: Hash functions (topic).

Let p be a 256-bit prime, and let E be an elliptic curve defined over $\mathbb{Z}_p$ with $|E(\mathbb{Z}_p)| = n$ a prime. Let $P, Q \in E(\mathbb{Z}_p)$ be randomly selected points, neither of which is the point at infinity; these points are fixed and public. Define the hash function $H : [0, n-1] \times [0, n-1] \to E(\mathbb{Z}_p)$ by $H((a,b)) = aP + bQ$. That is, messages are pairs (a,b) of integers in the interval $[0, n-1]$, and the hash of such a message is the elliptic curve point $aP + bQ$.

(a) Define what it means for H to be preimage resistant.

(b) Define what it means for H to be collision resistant.

(c) Prove, under a reasonable computational assumption (which you should state), that H is collision resistant.

(d) Prove, under a reasonable computational assumption (which you should state), that H is preimage resistant.

Problem 7/2020: Signature schemes (topic).

(a) Define what it means for a signature scheme to be secure.

In the basic (textbook) RSA signature scheme, the public key is a pair (n, e) and the secret key is a pair (n, d). The signature on a message $m \in \mathbb{Z}_n$ is $s \equiv m^d \pmod{n}$, and verification is done by checking if $s^e \equiv m \pmod{n}$.

(b) Show how to come up with some valid message/signature pairs under a key-only attack on the basic RSA signature scheme.

(c) Show how to come with a forgery for any given message m under a chosen-message attack on the basic RSA signature scheme, wherein the adversary is allowed to obtain signatures for at most two messages (neither of which is equal to m).

(d) One way of building a secure RSA-based signature scheme is to sign $H(m)$ using a "full-domain hash function H". The security proof for RSA-FDH is "in the random oracle model". Briefly explain what the random oracle model is.

Problems from the 2019 exam

Problem 1/2019: Elementary number theory (solution, topic).

Let $n = pq$ be an RSA modulus. Prove the following:

$$\prod_{x \in \mathbb{Z}_n^*} x \equiv 1 \pmod{n}.$$

Problem 2/2019: Hash functions (topic).

Recall that AES is a block cipher with message space $\{0,1\}^{128}$ and key space $\{0,1\}^{128}$. Let x and y denote bitstrings of length 128. Which of the following hash functions $H_i : \{0,1\}^{256} \to \{0,1\}^{128}$, if any, are preimage resistant? No justification is needed except that a lack of preimage resistance must be justified.

(i) $H_1(x,y) = \mathrm{AES}_x(y) \oplus y$.

(ii) $H_2(x,y) = \mathrm{AES}_y(x) \oplus x$.

(iii) $H_3(x,y) = \mathrm{AES}_y(x) \oplus y$.

Problem 3/2019: Symmetric-key encryption (topic).

Let E be a block cipher where both the block length and key size are 64 bits. Consider the "improved" block cipher whose encryption function is given by $m \mapsto c = E_k\big(m \otimes k_1\big)$ where m is the 64-bit plaintext, c is the 64-bit ciphertext, k_0 is a 64-bit string, k_1 is a 64-bit string, and the key is the 128-bit string (k_0, k_1).

(a) Give the decryption algorithm for the "improved" cipher.

(b) Describe how to break the "improved" cipher by brute force in time much less than 2^{128} using a known-plaintext attack.

Problem 4/2019: Cryptanalysis of RSA (topic).

Suppose that Alice and Bob agree to generate and share a common RSA modulus $n = pq$ for use in RSA encryption. Alice chooses an encryption exponent e_a and the corresponding decryption exponent d_a, and Bob chooses an encryption exponent e_b and the corresponding decryption exponent d_b. Suppose furthermore $\gcd(e_a, e_b) = 1$. Show that an adversary who acquires a pair of ciphertexts $c_a \equiv m^{e_a} \pmod{n}$ and $c_b \equiv m^{e_b} \pmod{n}$ corresponding to a single message m encrypted to Alice and Bob respectively can recover the plaintext message m efficiently.

Problem 5/2019: Elliptic curves (topic).

Let $E\colon y^2 = x^3 + ax + b$ be an elliptic curve defined over $\mathbb{Z}_p$, where $p > 3$ is prime.

(a) Prove the formula

$$\#E(\mathbb{Z}_p) = p + 1 + \sum_{x=0}^{p-1} \left(\frac{x^3 + ax + b}{p} \right),$$

where the expression inside the summation is the Legendre symbol.

(b) If furthermore $b = 0$ and $p \equiv 3 \pmod{4}$, show that $\#E(\mathbb{Z}_p) = p + 1$.

Problem 6/2019: Provable security (topic).

The Boneh–Franklin cryptosystem is defined as follows.

Public parameters: A cryptographic pairing $e : G \times G \to GT$, and two elements $g, h \in G$, each of which generates the cyclic group G.

Key generation: Choose $\alpha \in \mathbb{Z}$. The public key is g^α. The private key is h^α.

Encryption: Given $m \in \{0,1\}^k$, the encryption of m is $(g^r, e(g^\alpha, h^r) \oplus m)$ for random $r \in \mathbb{Z}$. Here k is the number of bits used to represent an element of GT.

Decryption: Given a ciphertext (c_1, c_2), the plaintext is $c_2 \oplus e(c_1, h^\alpha)$.

(a) Show that the Boneh–Franklin scheme is **OW-CPA** under the Bilinear Diffie–Hellman (BDH) assumption: Given g, g^α, g^β, and h, it is infeasible to compute $e(g^\alpha, h^\beta)$ from the ciphertext.

(b) Show that the Boneh–Franklin scheme is **IND-CPA** under the Decisional Bilinear Diffie–Hellman (DBDH) assumption.

$\mathrm{OW\text{-}CPA}^{\mathcal{A}}_{\mathrm{Enc}}$	
1:	$(\mathsf{pk}, \mathsf{sk}) \leftarrow_{\$} \mathsf{KGen}(1^\lambda)$
2:	$m \leftarrow_{\$} M$
3:	$c \leftarrow_{\$} \mathsf{Enc}(\mathsf{pk}, m)$
4:	$m' \leftarrow_{\$} \mathcal{A}(1^\lambda, \mathsf{pk}, c)$
5:	**return** $m \stackrel{?}{=} m'$

$\mathrm{IND\text{-}CPA}^{\mathcal{A}}_{\mathrm{Enc}}$	
1:	$(\mathsf{pk}, \mathsf{sk}) \leftarrow_{\$} \mathsf{KGen}(1^\lambda)$
2:	$b \leftarrow_{\$} \{0, 1\}$
3:	$(m_0, m_1) \leftarrow_{\$} \mathcal{A}(1^\lambda, \mathsf{pk})$
4:	$c \leftarrow_{\$} \mathsf{Enc}(\mathsf{pk}, m_b)$
5:	$b' \leftarrow_{\$} \mathcal{A}(1^\lambda, \mathsf{pk}, c)$
6:	**return** $b \stackrel{?}{=} b'$

Figure: OW-CPA and IND-CPA definitions.

Problems from the 2018 exam

Problem 1/2018: Block ciphers (topic).

Recall that DES is a block cipher with key space $K = \{0,1\}^{56}$, plaintext space $M = \{0,1\}^{64}$, and ciphertext space $C = \{0,1\}^{64}$.

(a) Let $\overline{m}$ denote the bitwise complement of a bit string m (i.e., $\overline{m} = m \oplus 11\ldots1$). By examining the description of DES, one can see that if $c = \mathrm{DES}_k(m)$ then $\overline{c} = \mathrm{DES}_{\overline{k}}(\overline{m})$. Can you use this property of DES to (slightly) improve the running time of exhaustive key search under a chosen-plaintext attack?

(b) Recall that Triple-DES has key space $K = \{0,1\}^{168}$. A plaintext $m \in \{0,1\}^{64}$ is encrypted under key $k = (k_1, k_2, k_3)$ (where $k_1, k_2, k_3 \in \{0,1\}^{56}$) as follows:

$$E_k(m) = DES_{k_3}(DES_{k_2}(DES_{k_1}(m))).$$

Describe a known-plaintext attack on Triple-DES that is significantly faster than exhaustive key search. Estimate the time and space requirements of your attack.

Problem 2/2018: Hash functions (topic).

Let q be a prime, and let G be a (multiplicatively written) group of order q. Let g and h be randomly selected elements from $G \setminus \{1\}$. Consider the function $H_{g,h} : \mathbb{Z}_q \times \mathbb{Z}_q \to G$ defined by $H_{g,h} : (x,y) \mapsto g^x h^y$. Henceforth we will denote $H_{g,h}$ by H.

(a) Show that for any $k \in G$, there are exactly q distinct solutions $(x,y) \in \mathbb{Z}_q \times \mathbb{Z}_q$ to the equation $g^x h^y = k$.

(b) Prove that if the discrete logarithm problem in G is intractable then H is collision resistant.

(c) Is H preimage resistant?

Problem 3/2018: Elementary number theory (solution, topic).

(a) Let $n \geq 3$ be an integer. Suppose that there exists an integer a such that $a^{n-1} \equiv 1 \pmod{n}$ and $a^{(n-1)/q} \not\equiv 1 \pmod{n}$ for all prime divisors q of $n-1$. Prove that n is prime.

(b) The *Fermat numbers* are $F_k = 2^{2^k} + 1$ for $k \geq 1$. Prove that for $k \geq 2$, F_k is prime if and only if $5^{(F_k-1)/2} \equiv -1 \pmod{F_k}$. (It may help to remember Euler's Theorem: If p is an odd prime, then $(\frac{a}{p}) \equiv a^{(p-1)/2} \pmod{p}$.)

Problem 4/2018: Number-theoretic algorithms (topic).

(a) An instance of CHREM (Chinese Remainder Problem) is a pair of distinct primes p and q, and two integers $a \in [0, p-1]$ and $b \in [0, q-1]$. The problem is to determine the unique integer $x \in [0, pq-1]$ such that $x \equiv a \pmod{p}$ and $x \equiv b \pmod{q}$. Design (and analyze) a polynomial-time algorithm for CHREM.

(b) Let $n = pq$, where p and q are distinct primes satisfying $p \equiv q \equiv 3 \pmod{4}$. Let FACTOR be the problem of factoring n. Let SQUARE-ROOT be the problem of finding one square root of a $\in \mathbb{Z}_n$ that is a quadratic residue modulo n. Prove that SQUARE-ROOT $\leq_P$ FACTOR.

Problem 5/2018: RSA (topic).

(a) Suppose that Alice's RSA public key is $(n = 143, e = 7)$. Determine her private key d.

(b) Let (n, e) be an RSA public key, where $n = pq$, and e is an integer with $1 < e < \phi(n)$ and $\gcd(e, \phi(n)) = 1$. It is known that the number of plaintexts $m \in [0, n-1]$ satisfying $m^e \equiv m \pmod{n}$ is

$$[1 + \gcd(e-1, p-1)] \cdot [1 + \gcd(e-1, q-1)].$$

Such a plaintext message m is called an *unconcealed message* since its RSA cyphertext is equal to m itself. Prove that there is at least one value of e, $1 < e < \phi(n)$, $\gcd(e, \phi(n)) = 1$, such that $m^e \equiv m \pmod{n}$ for all $m \in [0, n-1]$.

Problem 6/2018: Elliptic curves (topic).

Let p be an odd prime satisfying $p \equiv 2 \pmod{3}$. Consider the elliptic curve $E : Y^2 = X^3 + b$ defined over $\mathbb{F}_p$ with $b \neq 0$.

(a) Prove that the mapping $x \mapsto x^3$ is a bijection on $\mathbb{F}_p$.

(b) Prove that the number of points in $E(\mathbb{F}_p)$ is $p + 1$.

(c) Let $R = (x, y)$ be a point in $E(\mathbb{F}_p)$. Given y, explain how to compute x efficiently.

Problem 7/2018: ECDSA (topic).

Recall the ECDSA signature scheme. The domain parameters consist of a 256-bit prime p, an elliptic curve E defined over $\mathbb{Z}_p$ with prime $n = \#E(\mathbb{Z}_p)$, and a point $P \in E(\mathbb{Z}_p)$ with $P \neq \infty$. Alice's private key is $a \in_R [1, n-1]$ and her public key is $A = aP$. To sign a message $M \in \{0,1\}^*$, Alice does the following:

(i) Select a per-message secret $k \in_R [1, n-1]$.
(ii) Compute $m = \text{SHA256}(M)$.
(iii) Compute $R = kP$. Let $r = x(R) \bmod n$ and check that $r \neq 0$.
(iv) Compute $s = k^{-1}(m + ar) \bmod n$, and check that $s \neq 0$.
(v) Alice's signature on M is (r, s).

To verify Alice's signature (r, s) on M, Bob does the following:

(i) Obtain an authentic copy of Alice's public key A.
(ii) Check that $1 \leq r, s \leq n - 1$.
(iii) Compute $m = \text{SHA256}(M)$.
(iv) Compute $u_1 = ms^{-1} \bmod n$ and $u_2 = rs^{-1} \bmod n$.
(v) Compute $V = u_1 P + u_2 A$ and let $v = x(V) \bmod n$.
(vi) Accept if and only if $v = r$.

(a) Define what it means for a signature scheme to be secure.

(b) Suppose now that an adversary knows a message M such that $\text{SHA256}(M) = 0$. Show that the adversary can efficiently compute a valid signature for M. (The adversary knows the domain parameters and Alice's public key A, but does not have access to a signing oracle.)

Problems from the 2017 exam

Problem 1/2017: Symmetric-key encryption (topic).

The original DES block cipher is limited to a 56-bit key and 64-bit plaintext/ciphertext blocks. The DES-X block cipher, proposed by Ron Rivest, uses a 184-bit key (k, k_1, k_2) where $k \in \{0,1\}^{56}$ and $k_1, k_2 \in \{0,1\}^{64}$. The encryption of a plaintext $m \in \{0,1\}^{64}$ is given by

$$E(m) = \mathrm{DES}_k(m \oplus k_1) \oplus k_2,$$

where $\mathrm{DES}_k(m)$ denotes the DES-encryption of a 64-bit plaintext block m with 56-bit secret key k.

(a) Describe the decryption procedure.

(b) Suppose that the XOR with k_1 is omitted, i.e.

$$E(m) = \mathrm{DES}_k(m) \oplus k_2,$$

where the key (k, k_2) is now 120 bits. Describe a chosen-ciphertext attack that recovers the secret key using roughly 2^{56} DES operations.

Problem 2/2017: Hash functions (topic).

(a) Define what it means for a hash function to be *collision resistant*.

(b) Define what it means for a hash function to be *preimage resistant*.

(c) Define what it means for a hash function to be *second-preimage resistant*.

(d) Let $G\colon \{0,1\}^{2n} \to \{0,1\}^n$ and $H\colon \{0,1\}^{2n} \to \{0,1\}^n$ be two hash functions. Define the function $F\colon \{0,1\}^{2n} \times \{0,1\}^n \to \{0,1\}^n$ by $F(x,y) = H(G(x), G(x))$. (Here, the comma "," denotes concatenation.) Prove that if G and H are collision resistant, then F is also collision resistant.

(e) Suppose that $f\colon \{0,1\}^{n+r} \to \{0,1\}^n$ is a preimage resistant function. Define the function $H\colon \{0,1\}^{2(n+r)} \to \{0,1\}^n$ as follows. Given $x \in \{0,1\}^{2(n+r)}$, write $x = x_L \,\|\, x_R$ where $x_L, x_R \in \{0,1\}^{n+r}$. (Here, $\|$ denotes concatenation.) Then define

$$H(x) = f\big(x_L \oplus x_R\big).$$

Prove that H is not second-preimage resistant.

Problem 3/2017: Elementary number theory (incomplete solution, topic).

Note: Parts (a) and (b) are unrelated.

(a) Let p be a prime, $n \in \mathbb{N}$, and $q = p^n$. Prove that the finite field $\mathbb{F}_q$ has $q-2$ generators if and only if $q-1$ is a Mersenne prime.

(b) Let $m \geq 3$ be an integer. Prove that if a is a quadratic residue modulo m, and $ab \equiv 1 \pmod{m}$, then b is also a quadratic residue. Now let p be a prime of the form $p = 4k+3$. Prove that the product of all the quadratic residues modulo p is congruent to 1.

Problem 4/2017: Integer factorization (topic).

(a) Describe the *random squares method* for factoring a number n that is not a prime or a prime power. You are not expected to analyze the running time of the algorithm. (Note: In Stinson's book, the algorithm is called "Dixon's random squares algorithm". In Koblit's book, the algorithm is called "Factor base algorithm".)

(b) Explain the trade-off that dictates the optimal size of the factor base.

Problem 5/2017: RSA signatures (topic).

Recall that in the Full-Domain Hash (FDH) RSA signature scheme, an entity with public key (n, e) and private key d generates a signature s on a message m by computing

$$s \equiv H(m)^d \pmod{n}.$$

Here $H : \{0,1\}^* \to \{0, 1, \ldots, n-1\}$ is a hash function.

(a) Show that FDH RSA is insecure against passive adversaries if H is not preimage resistant.

(b) Prove that if finding e-th roots modulo n is intractable, and if H is a random function, then FDH RSA is existentially unforgeable by an adversary who can mount an adaptive chosen-message attack.

Problem 6/2017: Discrete logarithm and Diffie-Hellman problems (topic).

Let G be a group of prime order $n > 2$ generated by α. The notation $A \leq_P B$ means that problem A polynomial-time reduces to problem B.

(a) Recall that the discrete logarithm problem in G with respect to α (DLP_α) is the following: given $\gamma \in G$, find the integer $\ell \in [0, n-1]$ that satisfies $\gamma = \alpha^\ell$. Now, let β be another generator of G. Prove that $\text{DLP}_\alpha \leq_P \text{DLP}_\beta$. (This proves that the hardness of the DLP does not depend on the choice of generator.)

(b) Recall that the Diffie-Hellman Problem (DHP) is the following: given $\alpha^x, \alpha^y \in G$, compute α^{xy}. The problem INV is the following: given $\alpha^x \in G$, compute α^{x-1}. Prove that INV $\leq_P$ DHP.

(c) The problem SQUARE is the following: given $\alpha^x \in G$, compute α^{x^2}. Prove that DHP $\leq_P$ SQUARE.

Problem 7/2017: Elliptic curves (topic).

Let $E : Y^2 = X^3 + aX + b$ be an elliptic curve defined over $\mathbb{Z}_p$, where $p > 3$ is prime.

(a) Prove the formula

$$\#E(\mathbb{Z}_p) = p + 1 + \sum_{x=0}^{p-1} \left(\frac{x^3 + ax + b}{p} \right)$$

where the expression inside the summation is the Legendre symbol.

(b) Now suppose that $x^3 + ax + b$ splits into three distinct linear factors modulo p. Show that $E(\mathbb{Z}_p)$ is not cyclic.

Problems from the 2016 exam

Problem 1/2016: Hash functions (topic).

In the triangle of the three properties of a hash function:

- collision resistant
- preimage resistant
- second-preimage resistant

enter six symbols $\in \{\Rightarrow, \Leftrightarrow\}$ to indicate which property implies the other and which does not. Prove three out of the six directions.

Problem 2/2016: Elementary number theory (solution, topic).

Suppose that $p = 2^{2^k} + 1$ is prime, where $k \geq 1$. (a) Prove that any quadratic nonresidue modulo p is a generator of F_p^*. (b) Hence show that 7 is a generator of F_p^*.

Problem 3/2016: Elementary number theory (solution, topic).

Let n be an RSA modulus. Does n always, sometimes, or never have a primitive root? (Recall that a primitive root modulo n is an element of order $\phi(n)$ in the multiplicative group of units $\mathbb{Z}_n^*$.)

Problem 4/2016: RSA (topic).

Suppose that textbook RSA is used to encrypt a random 56-bit DES key k without padding; that is, the value of k as an integer is used as an RSA plaintext. Given the corresponding RSA ciphertext, give a (classical) algorithm that, with high probability, recovers the key k in substantially fewer than 2^{56} operations. Hint: Use the fact that a random integer 56-bit integer factors into a product of two integers less than 2^{29} with high probability.

Problem 5/2016: Discrete logarithm problem (topic).

Let $p = 2^k + 1$ be a prime number. Describe and analyze a polynomial-time algorithm for solving the discrete logarithm problem in $\mathbb{Z}_p^*$. (Recall that the DLP in $\mathbb{Z}_p^*$ is the following: given p, a generator g of $\mathbb{Z}_p^*$, and $h \in \mathbb{Z}_p^*$, find the integer $\ell \in [0, p-2]$ such that $h = g^\ell \bmod p$.)

Problem 6/2016: Message authentication codes (topic).

Recall the definition of CBC-MAC:

(a) Is CBC-MAC with one-block inputs existentially unforgeable under a chosen-message attack (EUF-CMA)? (b) Is CBC-MAC with variable-length inputs existentially unforgeable under a chosen-message attack (EUF-CMA)?

Problem 7/2016: Identification schemes (topic).

Let G be a cyclic group of prime order p with generator g. Suppose the verifier is given $\beta = g^\alpha$ for some randomly selected $\alpha \in \mathbb{Z}_p$. Consider the zero-knowledge proof of knowledge of α in Figure 1.

Figure 1: Zero-knowledge proof of knowledge of α

(a) Show that the proof is zero-knowledge for an honest verifier. (b) Show that a cheating prover can succeed with probability 1/2. (c) Describe how to modify the protocol so that the prover's cheating probability is reduced to negligible levels.

Problem 8/2016: Provable security (topic).

Consider the Zheng-Sebberry public-key encryption scheme (1993):

Public parameters: A cyclic group G, a generator g of G, and two random oracles

$$H_1 : \{0,1\}^t \to \{0,1\}^n, \qquad H_2 : G \to \{0,1\}^{t+n}.$$

Key generation: Choose a private key $x \in \mathbb{Z}$. The corresponding public key is $h = g^x$.

Encryption: To encrypt $m \in \{0,1\}^t$, choose $y \in \mathbb{Z}$ and compute

$$Y = g^y, \qquad c = H_2(h^y) \oplus (m \,\|\, H_1(m)).$$

The ciphertext is (Y, c).

Decryption: Compute $c \oplus H_2(Y^x)$. If the leftmost t bits of the result map to the rightmost n bits under H_1, then output the leftmost t bits; otherwise output NULL.

Show that the Zheng-Sebberry scheme is not IND-CCA2. (IND-CCA2 means "indistinguishable against adaptive chosen-ciphertext attack". In this attack, the adversary selects two plaintexts m_0, m_1, is then given the encryption c of m_b (where $b \in \{0, 1\}$), and has to determine b with probability significantly greater than $1/2$. The adversary is also given access to a decryption oracle to which it can present any ciphertext for decryption except for the challenge ciphertext c itself.)

Problems from the 2013 exam

Problem 1/2013: Hash functions (topic).

(a) Define what it means for a hash function to be preimage resistant.

(b) Let E be a block cipher with 80-bit keys and an 80-bit block size. Fix a publicly known 80-bit initialization vector IV. Define a hash function $H : \{0,1\}^{160} \to \{0,1\}^{80}$ as follows:

$$H(m) = E_{m_2}\big(E_{m_1}(\text{IV})\big),$$

where m_1 (respectively, m_2) represents the first (respectively, last) 80 bits of m. Give (and analyze) an algorithm to find preimages for H. Your algorithm should use no more than $\approx 2^{40}$ invocations of E.

Problem 2/2013: Block ciphers (topic).

Recall that the CBC block cipher mode of operation encrypts a message $m_1 m_2 \cdots m_n$ to the ciphertext $c_0 c_1 c_2 \cdots c_n$ where c_0 is chosen at random and

$$c_i = E_k(m_i \oplus c_{i-1}).$$

(a) Explain how decryption is performed with CBC.

(b) We define a new block cipher mode of operation known as UBC (Useless Block Chaining), with

$$c_i = E_k(m_i) \oplus c_{i-1}.$$

Show that UBC is, in fact, useless.

Problem 3/2013: Elementary number theory (incomplete solution, topic).

Let p denote an odd prime.

(a) Let q be a generator (a.k.a. a primitive root) modulo p. Show that q is not a quadratic residue modulo p.

(b) Suppose $y \equiv g^x \pmod{p}$ for some integer $0 < x < p-1$. Show how one can efficiently find the least significant bit of the binary expansion of x.

Problem 4/2013: Provable security (topic).

Let p be a large prime, and let q be a large prime divisor of $p-1$. Let q be an element of order q in $\mathbb{Z}_p^*$, and let G denote the subgroup of $\mathbb{Z}_p^*$ generated by g. We consider the ElGamal encryption scheme in the group G, which is defined as follows:

Setup: Public parameters p, q, g are chosen.

Key generation: Choose $x \in \mathbb{Z}_q^*$ at random. The public key is g^x and the private key is x.

Encryption: The message $m \in G$ and a public key g^x, choose a random integer r and output the ciphertext $c = (g^x, mg^{xr})$.

Decryption: Given a ciphertext $c = (\rho, \sigma)$, output $m = \sigma/\rho^x$.

(a) It is conjectured that the Decision Diffie-Hellman (DDH) assumption holds in the group G. Assuming this conjecture, prove that the ElGamal encryption scheme is IND-CPA secure. (Recall that IND-CPA means "indistinguishable against chosen-plaintext attacks".)

(b) Suppose that, in the encryption function defined above, the message space is taken to be $\mathbb{Z}_p^*$ instead of G. Show that the resulting encryption scheme is not IND-CPA secure.

Problem 5/2013: RSA (topic).

Alice has a corrupted copy of Bob's RSA public key (n, e), in which one bit of the public exponent e is wrong. Alice encrypts a message m under the textbook RSA scheme using this corrupted public key, and sends the resulting ciphertext c_1 to Bob. Later, Alice obtains a correct copy of Bob's RSA public key, and sends an encryption c_2 of the same message m under textbook RSA using the correct key. An adversary, who knows Bob's public key, obtains both c_1 and c_2. Show how the adversary can obtain m.

Problem 6/2013: Elliptic curves (topic).

Let p be an odd prime satisfying $p \equiv 2 \pmod 3$. Consider the elliptic curve $E : y^2 = x^3 + b$ defined over $\mathbb{F}_p$ ($b \neq 0$).

(a) Prove that the mapping $x \mapsto x^3$ is a bijection on $\mathbb{F}_p$.

(b) Prove that the number of points in $E(\mathbb{F}_p)$ is $p + 1$.

(c) Let $R = (x_R, y_R)$ be a point on $E(\mathbb{F}_p)$. Given y_R, explain how to compute x_R efficiently.

Problems from the 2011 exam

Problem 1/2011: Hash functions (topic).

(a) Give an example of a hash function $H : \{0,1\}^* \to \{0,1\}^*$ that is collision resistant but not preimage resistant. Justify your choice!

(b) Define what it means for a pair of permutations f_0, f_1 on S to be claw-free.

(c) Let $p, q \equiv 3 \pmod 4$, and let $n = pq$. Let S denote the set of squares modulo n, coprime with n. Let $a_0, a_1 \in S$, and define the functions $f_0 : S \to S$ and $f_1 : S \to S$ by

$$f_0(x) = a_0 x^2 \pmod n, \qquad f_1(x) = a_1 x^2 \pmod n.$$

Then f_0 and f_1 are permutations on S (you do not need to show this). Show that under the assumption that factoring n is computationally infeasible, f_0 and f_1 is a claw-free pair of permutations.

(d) Using f_0 and f_1, construct a collision-free hash function

$$H : \{0,1\}^* \to S.$$

Problem 2/2011: Block ciphers (topic).

Recall that a Feistel cipher has a round function of the following form

$$L_i^t = R_i^{t-1}, \qquad R_i^t = L_i^{t-1} \oplus f(R_i^{t-1}, K_i).$$

The plaintext is $L^0 \,||\, R^0$ and the ciphertext is $L^n \,||\, R^n$.

(a) What properties does f need to satisfy in order for encryption to be invertible? Justify.

(b) Describe the decryption algorithm.

Problem 3/2011: Elementary number theory (topic).

Throughout this question, n is an odd integer, and a is an integer, $1 < a < n$.

(a) Assume n is prime, and write $n - 1 = 2^k m$ with m odd. Assume $a^{2^i m} \not\equiv -1 \pmod{n}$ for all $0 \leq i < k$. Prove that $a^{2^i m} \equiv 1 \pmod{n}$ for all $0 \leq i \leq k$.

(b) Let n be prime. Prove that $\left(\frac{a}{n}\right) \equiv a^{(n-1)/2} \pmod{n}$.

(c) Now let n be composite, again write $n - 1 = 2^k m$ with m odd. Assume $\gcd(a, n) = 1$. Recall that n is called a strong pseudoprime to the base a if

$$a^n \equiv 1 \pmod{n} \quad \text{or} \quad \text{there exists } 0 \leq i < k \text{ such that } a^{2^i m} \equiv -1 \pmod{n}.$$

Further recall that n is called an Euler pseudoprime to the base a if

$$a^{(n-1)/2} \equiv \left(\frac{a}{n}\right) \pmod{n}.$$

Prove that for $n \equiv 3 \pmod{4}$, n is a strong pseudoprime to the base a if and only if n is an Euler pseudoprime to the base a.

Problem 4/2011: Provable security (topic).

The Blum-Goldwasser public key cryptosystem is given as follows.

- Key generation: Choose distinct primes p and q congruent to 3 (mod 4). The public key is $n = pq$ and the private key is (p, q). - Encryption: A message m consists of a single bit. To encrypt m, choose a random $x \in (\mathbb{Z}/n\mathbb{Z})^{\times 2}$ (the quadratic residues modulo n) and compute

$$i)\ b = \mathrm{LSB}(x), \quad ii)\ c = b \oplus m, \quad iii)\ y = x^2 \pmod{n}.$$

The ciphertext is (c, y).

Prove that the cryptosystem is IND-CPA. State any necessary assumptions.

Problem 5/2011: Pseudo-random bit generators (topic).

Let (n, e) be an RSA public key, and let d be the corresponding RSA private key. Let $f : \mathbb{Z}_n \to \mathbb{Z}_n$ be defined by $f(x) = x^e \bmod n$. Let

$$B(x) = \begin{cases} 1, & \text{if } x^d \bmod n \text{ is odd,} \\ 0, & \text{if } x^d \bmod n \text{ is even.} \end{cases}$$

Clearly, $B(x)$ is easy to compute given x and $f^{-1}(x)$ (since $f^{-1}(x) = x^d \bmod n$). It can be shown (and you may assume) that $B(x)$ is hard to compute given only x.

(a) Define (informally) what it means for a PRBG G to pass the next bit test.

(b) Describe a cryptographically secure PRBG G which takes as seed random $x_0 \in \mathbb{Z}_n$ and uses f and B above.

(c) Prove that the generator G passes the next bit test.

Problem 6/2011: Elliptic curves (topic).

Consider the Kobitz (Koblitz) curve defined over $\mathbb{F}_2$:

$$E: \ y^2 + xy = x^3 + x^2 + 1.$$

(a) Show that $\#E(\mathbb{F}_2^r)$ is even for all $r \geq 1$.

(b) Show that if $r > 4$ and $\#E(\mathbb{F}_{2^r}) = 2p$ where p is a prime number, then r must be prime.

Problems from the 2007 exam

Problem 1/2007: Hash functions (topic).

(a) Is a collision-resistant hash function necessarily preimage resistant?

(b) Let (n, e) be an RSA public key, where n is 2048 bits in length. The corresponding RSA private key is not known to anyone. Define a hash function $H : \{0,1\}^* \to [0, n-1]$ as follows: $H(m) = m^e \bmod n$, where m denotes the integer whose binary representation is m. Is H collision resistant? (Justify your answer.)

(c) Let $IV \in \{0,1\}^n$ be a fixed initialization vector, and let $f\colon \{0,1\}^{n+r} \to \{0,1\}^n$ be a compression function. Define the hash function H as follows: to hash a message x of bitlength $b < 2^r$, the message is first divided into r-bit blocks: $\bar{x} = x_1, x_2, \ldots, x_t$ (where the last block is padded with 0 bits if necessary). Define x_{t+1} to be the r-bit binary representation of b. Then $H_0 = \text{IV}$ and $H_i = f(H_{i-1}, x_i)$ for $i = 1, 2, \ldots, t$. Then $H(x)$ is defined to be H_{t+1}. Prove that if f is collision resistant, then H is also collision resistant.

Problem 2/2007: Elementary number theory (solution, topic).

(a) Let p be an odd prime. Prove that a quadratic residue modulo p can never be a generator of $\mathbb{F}_p^*$.

(b) Let p be an odd prime. Prove that -3 is a quadratic residue modulo p if and only if $p \equiv 1 \pmod 3$.

(c) Let $p > 3$ be a Mersenne prime. Prove that 3 is a quadratic nonresidue modulo p.

Problem 3/2007: Discrete logarithm problem (topic).

Let $p = 2^k + 1$ be a prime number. Describe and analyze a polynomial-time algorithm for solving the discrete logarithm problem in $\mathbb{F}_p^*$. (Recall that the DLP in $\mathbb{F}_p^*$ is the following: given p, a generator α of $\mathbb{F}_p^*$, and $\beta \in \mathbb{F}_p^*$, find the integer $l \in [0, p-2]$ such that $\beta \equiv \alpha^l \pmod p$.)

Problem 4/2007: Security of the basic ElGamal public-key encryption scheme (topic).

Let G be a (cyclic) group of prime order $n > 2$, and let α be a generator of G. We assume that the group operation in G can be computed in polynomial time. Recall that the Diffie-Hellman problem for G with respect to α (DHP_α) is the following: given α^a and α^b, compute α^{ab}. The Decisional Diffie-Hellman problem for G with respect to α (DDHP_α) is the following: given α^a, α^b, and α^c, decide whether $c \equiv ab \pmod{n}$.

In the basic ElGamal public-key encryption scheme, Alice's private key is an integer $a \in [1, n-1]$, and her public key is $\beta = \alpha^a$. To encrypt a plaintext message $m \in G$ for Alice, Bob selects $k \in_R [1, n-1]$, and sends the ciphertext $C = (\alpha^k, m \cdot \beta^k)$ to Alice.

In the following, we consider ciphertext-only attacks on the basic ElGamal public-key encryption scheme. The attacker has knowledge of the group parameters, Alice's public key β, and one or more ciphertexts.

(a) The ElGamal-decrypt problem is the following: Given a public key β and a ciphertext C, compute the corresponding plaintext. Prove that the ElGamal-decrypt problem is polynomial-time equivalent to DHP_α.

(b) Prove that the semantic security of the basic ElGamal public-key encryption scheme (under ciphertext-only attack) is polynomial-time equivalent to DDHP_α.

(c) Is the basic ElGamal public-key encryption scheme semantically secure against chosen-plaintext attacks? (Justify your answer.)

Problem 5/2007: Weil pairing (topic).

Let E be an elliptic curve over a field, and suppose $P, Q \in E[n]$ for some $n > 0$.

(a) Give the formula for the Weil pairing $e(P, Q)$ of P and Q.

(b) Prove that the choice of divisors in the formula does not affect the value of $e(P, Q)$. You may assume Weil reciprocity.

(c) Prove that the Weil pairing is bilinear.

Problem 6/2007: Embedding degree (topic).

Let E be an elliptic curve defined over $\mathbb{F}_q$, and denote by n the largest prime factor of $\#E(\mathbb{F}_q)$. Recall that the embedding degree of E is by definition equal to the multiplicative order of q in $(\mathbb{Z}/n\mathbb{Z})^*$.

(a) Let k be an integer. Prove that the embedding degree of E divides k if and only if $n \mid (q^k - 1)$, where t is the trace of E.

(b) Suppose $q = p^2$ and $t = p$ where p is prime. Determine the embedding degree of E.

Problems from the 2004 exam

Problem 1/2004: Hash functions (topic).

(a) Define what it means for a hash function to be collision resistant. (b) Define what it means for a hash function to be second preimage resistant. (c) Let E denote the family of encryption functions for the AES block cipher where plaintext blocks, ciphertext blocks, and keys are each 128 bits in length. Define a hash function $H : \{0,1\}^{128} \to \{0,1\}^{128}$ by $H(x,y) = E_x(y)$. Here, x and y are 128-bit blocks, and $E_x(y)$ denotes the encryption of the plaintext block y using the key x. Is H collision resistant? (Justify your answer.) (d) Is the hash function in (c) second preimage resistant? (Justify your answer.)

Problem 2/2004: Elementary number theory (topic).

Let p be an odd prime and let $n > 1$ be a positive integer. Recall that the multiplicative group $\mathbb{Z}_{p^n}^*$ is cyclic. Suppose that the integer g is a generator of $\mathbb{Z}_p^*$, and let $h = (p+1)g$. Prove that at least one of g or h is a generator of $\mathbb{Z}_{p^n}^*$. Hint: First prove that at least one of g^{p-1} or h^{p-1} can be expressed in the form $1 + kp$ where k is an integer that is not divisible by p.

Problem 3/2004: Bit security of the discrete logarithm problem (topic).

Let p be an odd prime, and let g be a generator of $\mathbb{Z}_p^*$. Consider the following three problems: DLP: Given p, g, and $x \in \mathbb{Z}_p^*$, determine the integer $a \in [0, p-2]$ such that $x \equiv g^a \pmod{p}$. (We write $a = \log_g x$.) DLP-LSB: Given p, g, and $x \in \mathbb{Z}_p^*$, determine $A(x)$ where

$$A(x) = \begin{cases} 1, & \text{if } \log_g x \text{ is even,} \\ 0, & \text{if } \log_g x \text{ is odd.} \end{cases}$$

DLP-MSB: Given p, g, and $x \in \mathbb{Z}_p^*$, determine $B(x)$ where

$$B(x) = \begin{cases} 1, & 0 \le \log_g x < \frac{p-1}{2}, \\ 0, & \frac{p-1}{2} \le \log_g x \le p-2. \end{cases}$$

(a) Prove that DLP $\leq_p$ DLP. (Recall that A $\leq_p$ B means that problem A polynomial-time reduces to problem B.) (b) Prove that DLP $\leq_p$ DLP-MSB. (c) Does DLP $\leq_p$ DLP-LSB? (Justify your answer.)

Problem 4/2004: Fault analysis attack on the RSA signature scheme (topic).

Suppose that a smart card is using the Chinese Remainder Theorem for RSA signature generation. That is, if (n, e) is the RSA public key and d is the corresponding private key, then signing a message m is performed as follows:

i) Compute $M = H(m)$.

ii) Compute $s_p \equiv M^{d_p} \pmod{p}$ and $s_q \equiv M^{d_q} \pmod{q}$, where $d_p = d \bmod (p-1)$ and $d_q = d \bmod (q-1)$.

iii) Find s, $0 \leq s \leq n-1$, such that

$$s \equiv s_p \pmod{p}, \qquad s \equiv s_q \pmod{q}.$$

(a) Prove that s is the correct signature of m (that is, prove that $s \equiv H(m)^d \pmod{n}$). (b) Explain why it might be advantageous to compute s using the procedure described above instead of computing $s = H(m)^d \bmod n$ directly using the repeated square-and-multiply algorithm. (c) Suppose now that an adversary can somehow induce the smart card to compute s_p incorrectly (and s_q correctly) while signing a message. Let s' be a resulting (incorrect) signature on m. Suppose that the adversary has access to the public key (n, e) and also the signed message (m, s'). Show how the adversary can efficiently factor n. (d) Suggest a (realistic and practical) method for preventing this attack.

Problem 5/2004: Provable security (topic).

Recall that in the Full-Domain Hash (FDH) RSA signature scheme, an entity with public key (n, e) and private key d generates a signature s on a message m by computing $s = H(m)^d \bmod n$. Here $H : \{0,1\}^* \to \{0,1\}^n$ is a hash function. Prove that if finding the roots mod n is intractable, and if H is a random function, then FDH RSA is existentially unforgeable by an adversary who can mount an adaptive chosen-message attack.

Problem 6/2004: Elliptic curves and finite fields (topic).

(a) Recall that the Trace function $\mathrm{Tr} : \mathbb{F}_{2^m} \to \mathbb{F}_2$ is defined by $\mathrm{Tr}(\alpha) = \sum_{i=0}^{m-1} \alpha^{2^i}$. Prove that exactly half of the elements in $\mathbb{F}_{2^m}$ have trace 0 (and the other half have trace 1). (b) Let $\alpha \in \mathbb{F}_{2^m}$. Prove that the equation $x^2 + x = \alpha$ has a solution $x \in \mathbb{F}_{2^m}$ if and only if $\mathrm{Tr}(\alpha) = 0$. (c) Let $E : \ y^2 + y = x^3$ be an elliptic curve over $\mathbb{F}_{2^m}$ where m is an odd positive integer. Prove that $|E(\mathbb{F}_{2^m})| = 2^m + 1$.

Problems from the 2003 exam

Problem 1/2003: Chosen-plaintext attack on two-key Triple-DES (topic).

Recall that DES is a block cipher with key space $K = \{0,1\}^{56}$, plaintext space $M = \{0,1\}^{64}$, and ciphertext space $C = \{0,1\}^{64}$. Encryption for *two-key Triple-DES* is defined as follows:

$$E_k(m) = \mathrm{DES}_{k_1}(\mathrm{DES}^{-1}_{k_2}(\mathrm{DES}_{k_1}(m))),$$

where $k = (k_1, k_2)$, and $k_1, k_2 \in_R \{0,1\}^{56}$ is the secret key. Design a *chosen-plaintext* attack on two-key Triple-DES that takes *roughly* 2^{56} steps. (A step is a DES encryption or decryption operation.) Provide an explanation of why your attack works, and a careful estimate of its space and time requirements. (Hint: your attack may need a *lot* of chosen plaintext/ciphertext pairs.)

Problem 2/2003: Elementary number theory (solution, topic).

(a) Prove that if $p = 2^m + 1$ is prime, then m is a power of 2.

(b) Suppose that $p = 2^m + 1$ is prime. Prove that any quadratic nonresidue is a generator of $\mathbb{F}_p^*$.

(c) Suppose that $p = 2^m + 1$ is prime. Prove that 5 is a generator of $\mathbb{F}_p^*$, except in the case $p = 5$.

Problem 3/2003: Partial key-exposure in RSA (topic).

This exercise shows that if the encryption exponent in RSA is $e = 3$, then the left half of the bits of d can be very easily computed. (More precisely, the possible values for the left half of the bits can be narrowed to one or two.)

Let $n = pq$ where p and q are primes with $5 \le p < q < 2p$. Let integers e and d satisfy $1 < e, d < \phi(n)$ and $ed \equiv 1 \pmod{\phi(n)}$.

(a) Prove that there exists an integer k satisfying $ed - k\phi(n) = 1$ and $1 \le k < e$.

(b) Let $\hat{d} = \lfloor \frac{kn+1}{e} \rfloor$. Prove that $|\hat{d} - d| < 3\sqrt{n}$.

(c) Prove that if $e = 3$ then $k = 2$.

Problem 4/2003: Diffie-Hellman problem (topic).

This exercise shows that the hardness of the Diffie-Hellman problem does not depend on the choice of generator.

Let G be a (cyclic) group of prime order $n > 2$, and let α be a generator of G. We assume that the group operation in G can be computed in polynomial time. Recall that the Diffie-Hellman problem for G with respect to α (DHP_α) is the following: given α^a and α^b, compute α^{ab}. In this question, you are given a polynomial-time algorithm A which solves DHP_α.

(a) Devise a polynomial-time algorithm which on input α^a and a positive integer k, outputs α^{a^k}.

(b) Devise a polynomial-time algorithm which on input α^a (with $a \not\equiv 1 \pmod{n}$), outputs $\alpha^{a^{-1}}$.

(c) Let β be a generator of G. Devise a polynomial-time algorithm for solving DHP_β (i.e., given β^a and β^b, compute β^{ab}).

Problem 5/2003: Security of the basic ElGamal public-key encryption scheme (topic).

Let G be a (cyclic) group of prime order $n > 2$, and let α be a generator of G. We assume that the group operation in G can be computed in polynomial time. Recall that the Diffie-Hellman problem for G with respect to α (DHP_α) is the following: given α^a and α^b, compute α^{ab}. The decision Diffie-Hellman problem for G with respect to α (DDHP_α) is the following: given α^a, α^b and α^c, decide whether $c \equiv ab \pmod{n}$.

In the basic ElGamal public-key encryption scheme, Alice's private key is an integer $a \in [1, n-1]$, and her public key is $\beta = \alpha^a$. To encrypt a plaintext message $m \in G$ for Alice, Bob selects $k \in_R [1, n-1]$, and sends the ciphertext $C = (\alpha^k, m\beta^k)$ to Alice.

In the following, we consider ciphertext-only attacks on the basic ElGamal public-key encryption scheme. The attacker has knowledge of the group parameters, Alice's public key β, and one or more ciphertexts.

(a) The ElGamal-decrypt problem is the following: Given a public key β and a ciphertext C, compute the corresponding plaintext. Prove that the ElGamal-decrypt problem is polynomial-time equivalent to DHP_α.

(b) Prove that the semantic security of the basic ElGamal public-key encryption scheme (under ciphertext-only attack) is polynomial-time equivalent to DDHP_α.

(c) Is the basic ElGamal public-key encryption scheme semantically secure against chosen-ciphertext attacks? (Justify your answer.)

Problem 6/2003: Elliptic curves (topic).

Let q be a power of an odd prime, $q \equiv 2 \pmod 3$.

(a) Prove that the mapping $x \mapsto x^3$ is a 1-1 map of $\mathbb{F}_q$ to itself.

(b) Consider the elliptic curve $E\colon y^2 = x^3 + b$ defined over $\mathbb{F}_q$. Prove that the number of points in $E(\mathbb{F}_q)$ is $q + 1$.

(c) Prove that $E(\mathbb{F}_q)$ is cyclic.

Problems from the 2002 exam

Problem 1/2002: Attempts to strengthen DES against exhaustive key search attacks (topic).

Recall that DES is a symmetric-key encryption scheme with a 56-bit key, and 64-bit plaintext and ciphertext blocks. Consider the following proposal for a new symmetric-key encryption scheme based on DES. The secret key for the new scheme is $k = (k_1, k_2)$, where $k_1 \in \{0,1\}^{56}$ and $k_2 \in \{0,1\}^{64}$ (so k is a 120-bit key). Let $m \in \{0,1\}^{64}$ be a plaintext message. Then encryption is defined as follows:

$$E_k(m) = \mathrm{DES}_{k_1}(m \oplus k_2).$$

(a) Show how this encryption scheme can be totally broken—that is, the secret key k can be recovered—by a known-plaintext attack using *roughly* 2^{56} DES encryption/decryption operations. Your attack should have little space requirements. You may assume that you have a moderate number of plaintext-ciphertext pairs $(m_i, c_i = E_k(m_i))$. Briefly justify why the number of such pairs you use is sufficient to uniquely determine the key with high probability.

(b) Is the encryption scheme with encryption function $E_k(m) = \mathrm{DES}_{k_1}(m \oplus k_2) \oplus k_2$ any more secure than DES? [Briefly justify your answer.]

(c) Is the encryption scheme with encryption function $E_k(m) = \mathrm{DES}_{k_1}(m \oplus k_2) \oplus k_3$ any more secure than DES? (Here, $k = (k_1, k_2, k_3)$ where $k_1 \in \{0,1\}^{56}$ and $k_2, k_3 \in \{0,1\}^{64}$.) [Briefly justify your answer.]

Problem 2/2002: Hash functions (topic).

Suppose $h_1 \colon \{0,1\}^{2m} \to \{0,1\}^m$ is a collision resistant hash function.

(a) Define $h_2 \colon \{0,1\}^{4m} \to \{0,1\}^m$ as follows:

(i) Write $x \in \{0,1\}^{4m}$ as $x = x_1 \parallel x_2$, where $x_1, x_2 \in \{0,1\}^{2m}$.

(ii) Define $h_2(x) = h_1(x_1) \parallel h_1(x_2)$.

Prove that h_2 is collision resistant.

(b) For an integer $i \geq 2$, define a hash function $h_i \colon \{0,1\}^{2^i m} \to \{0,1\}^m$ recursively from h_{i-1} as follows:

(i) Write $x \in \{0,1\}^{2^i m}$ as $x = x_1 \parallel x_2$, where $x_1, x_2 \in \{0,1\}^{2^{i-1} m}$.

(ii) Define $h_i(x) = h_{i-1}(x_1) \parallel h_{i-1}(x_2)$.

Prove that h_i is collision resistant.

(c) Let γ be a quadratic residue modulo p. Show that the two square roots of γ modulo p are $\pm\gamma^{(p+1)/4}$.

(d) Show that $L_1(\beta)$ can be efficiently computed given p, α, β. Here, $L_1(\beta)$ denotes the least significant bit of the discrete logarithm a of $\beta = \alpha^a \pmod{p}$ (i.e., $L_1(\beta) = 0$ if a is even, and $L_1(\beta) = 1$ if a is odd). Let $L_2(\beta)$ denote the second least significant bit of a. That is, $L_2(\beta) = 0$ if $a \equiv 0$ or $1 \pmod{4}$, and $L_2(\beta) = 1$ if $a \equiv 2$ or $3 \pmod{4}$.

Problem 3/2002: Carmichael numbers (solution, topic).

Assume throughout this question that n is square-free (i.e., n is not divisible by the square of a prime). Then $n = p_1 \cdots p_\ell$, where $p_1, \ldots, p_\ell$ are distinct primes. Such an integer n is a Carmichael number if $a^{n-1} \equiv 1 \pmod{n}$ for all integers a that are relatively prime to n. You may use the following facts in your solutions: (i) $a^{n-1} \equiv 1 \pmod{p_i}$ for all i implies $a^{n-1} \equiv 1 \pmod{n}$; (ii) for every prime p, there exists a primitive element mod p.

(a) Suppose that n is a Carmichael number. Prove that $p_i - 1$ divides $n - 1$ for all $i, 1 \leq i \leq \ell$.

(b) Suppose that $p_i - 1$ divides $n - 1$ for all $i, 1 \leq i \leq \ell$. Then prove that n is a Carmichael number.

(c) Prove that 561 is a Carmichael number.

Problem 4/2002: Speeding up RSA decryption (topic).

Let (n, e) be Alice's RSA public key, and let d be her corresponding private key. Recall that the RSA encryption operation is $c = m^e \bmod n$, while the RSA decryption operation is $m = c^d \bmod n$. One way to speed up RSA decryption is to precompute $d_p = d \bmod (p-1)$ and $d_q = d \bmod (q-1)$. Then decryption of a ciphertext c can be performed by computing $m_p = c^{d_p} \bmod p$ and $m_q = c^{d_q} \bmod q$, and then finding m, $0 \leq m \leq n-1$, such that $m \equiv m_p \pmod{p}$ and $m \equiv m_q \pmod{q}$.

(a) Describe a procedure (i.e., a formula) that Alice can use to compute m efficiently, given m_p and m_q.

(b) Prove that m is the correct decryption of c. That is, prove that $m \equiv c^d \pmod{n}$.

(c) Briefly justify the assertion that this method of decryption can speed up RSA decryption by approximately 75

(d) Devise an algorithm which, on input n and e, factors n in $O(\min(d_p, d_q))$ steps. (A "step" is any operation whose running time is polynomial in $\log n$.) State any assumptions you may make. [This exercise shows that Alice should not try to speed up decryption by selecting d so that d_p and d_q are too small.]

Problem 5/2002: Bit security of the discrete logarithm problem (topic).

Let p be a prime with $p \equiv 3 \pmod 4$. Let $\alpha \in \mathbb{Z}_p^*$ be a generator of $\mathbb{Z}_p^*$. The discrete logarithm problem in $\mathbb{Z}_p^*$ is the following: given α and $\beta \in \mathbb{Z}_p^*$, find the integer a, $0 \leq a \leq p-2$, such that $\beta = \alpha^a \pmod p$. Let $L_1(\beta)$ denote the least significant bit of a. That is, $L_1(\beta) = 0$ if a is even, and $L_1(\beta) = 1$ if a is odd. Let $L_2(\beta)$ denote the second least significant bit of a. That is, $L_2(\beta) = 0$ if $a \equiv 0$ or $1 \pmod 4$, and $L_2(\beta) = 1$ if $a \equiv 2$ or $3 \pmod 4$.

(a) Let $\gamma \in \mathbb{Z}_p^*$ be a quadratic residue modulo p. Show that the two square roots of γ modulo p are $\pm\gamma^{(p+1)/4}$.

(b) Show that $L_1(\beta)$ can be efficiently computed given p, α, β.

(c) Prove that $L_1(\beta) \neq L_1(\beta^{-1})$.

(d) Suppose that you have an efficient algorithm A (an oracle) for computing $L_2(\beta)$ given p, α, β. Use A to devise an efficient algorithm for computing $L_1(\beta)$ and hence break the discrete logarithm problem. Briefly justify that your algorithm is correct and efficient.

Problem 6/2002: Hash functions and DSA (topic).

We recall the DSA signature scheme. The system parameters consist of a 1024-bit prime p, a 160-bit prime divisor q of $p-1$, and an element $g \in \mathbb{Z}_p^*$ of order q. SHA-1 is a 160-bit hash function. Alice's private key is $a \in \mathbb{Z}_p^*$, while her public key is $h = g^a \bmod p$. To sign a message $M \in \{0,1\}^*$, Alice does the following:

1) Choose a per-message secret number k with $1 \leq k \leq q-1$ and compute $r \equiv (g^k \bmod p) \bmod q$. If $r = 0$, choose another k.

2) Compute $s \equiv k^{-1}(\text{SHA}-1(M) + ar) \pmod{q}$. If $s = 0$, choose another k.

The signature on M is the pair (r, s).

To verify a signature (r, s) on M, Bob does the following:

(i) Obtain an authentic copy of Alice's public key h.

(ii) Compute $m = \text{SHA}-1(M)$.

(iii) Check that $1 \leq r, s \leq q-1$.

(iv) Compute $u_1 = ms^{-1} \bmod q$ and $u_2 = rs^{-1} \bmod q$.

(v) Accept if $r \equiv (g^{u_1}h^{u_2}) \bmod p \bmod q$. (Otherwise reject.)

Recall that a signature scheme is secure if it is existentially unforgeable by chosen-message attacks. It is insecure if it is not secure.

(a) Define what it means for SHA-1 to be preimage resistant.

(b) Define what it means for SHA-1 to be 2nd preimage resistant.

(c) Define what it means for SHA-1 to be collision resistant.

(d) Prove that DSA is insecure if SHA-1 is not preimage resistant.

(e) Prove that DSA is insecure if SHA-1 is not 2nd preimage resistant.

(f) Prove that DSA is insecure if SHA-1 is not collision resistant.

Problem 7/2002: Duplicate signatures (topic).

The Elliptic Curve Digital Signature Algorithm (ECDSA) is as follows: Let p be a prime, and let E be an elliptic curve defined over $\mathbb{F}_p$. Let A be a point on E having prime order q, such that the Discrete Logarithm problem in $\langle A \rangle$ is infeasible. Let $\mathcal{P} = \{0,1\}^*$ denote the set of messages. Define the public parameters $\mathcal{K} = \{(p, q, E, A, m, B) : B = mA\}$, where $0 \leq m \leq q-1$. The values p, q, E, A and B are the public key, and m is the private key.

The signature for a message $x \in \mathcal{P}$ is computed as follows: For $K = (p, q, E, A, m, B)$, and for a (secret) random number k, $1 \leq k \leq q-1$, define

$$sig_K(x, k) = (r, s),$$

where $kA = (u, v)$ and $r = u \bmod q$, $s = k^{-1}(\text{SHA}-1(x) + mr) \bmod q$.

(If either $r = 0$ or $s = 0$, a new random value of k should be chosen.)

To verify A's signature (r, s) on M, Bob does the following:

(i) Obtain an authentic copy of Alice's public key h.

(ii) Compute $m = \text{SHA} - 1(M)$.

(iii) Check that $1 \leq r, s \leq q - 1$.

(iv) Compute $u_1 = ms^{-1} \bmod q$ and $u_2 = rs^{-1} \bmod q$.

(v) Accept iff $r = (hu_1 + Bu_2) \restriction_q$ (i.e., the x-coordinate modulo q of the point $u_1 A + u_2 B$).

Recall that a signature scheme is secure if it is existentially forgeable, etc.

(a) Suppose that two messages x and x' are signed with the same k under the same key, producing signatures (r, s) and (r', s'). Show that if SHA-1 is used as the hash, then $x \neq x'$ can still yield the same signature (r, s) with non-negligible probability. Explain how this leads to an attack on the ECDSA signature scheme.

(b) Suppose that two messages x and x' are said to have duplicate ECDSA signatures if SHA-1$(x) \neq$ SHA-1(x') but $\text{sig}_K(x, k) = \text{sig}_K(x', k')$ for some integers k, k' for some key parameters. Suppose the public key parameters p, q, E and A are fixed. Given any two messages x and x' such that SHA-1$(x) \neq$ SHA-1(x'), show that it is possible for Alice to choose a private key m (and hence a corresponding public key B) so that x and x' have duplicate ECDSA signatures under the key p, q, E, A, m and B.

(c) Suppose that Alice, say, signs message x with signature (r, s), and then later claims that she really signed the message x', where (r, s) is also a signature on x'. Show that an adversary can now easily compute Alice's secret key.

(d) We have shown that Alice can choose her private key in such a way that she can later construct duplicate signatures on two messages x and x'. Does this property mean that the ECDSA is "insecure"? (Discuss.)